A JAX–PETSc Toolset for Nonlinear Finite-Element Energy Minimization:

Local Automatic Differentiation, Sparse Assembly, and Solver Policy

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Abstract

Large nonlinear finite-element energy problems require reliable local derivatives, robust nonlinear globalization, and sparse distributed linear algebra that is matched to the mechanics of the problem. We develop a scientific toolset for studying these requirements as coupled numerical choices in nonlinear finite-element energy minimization and energy-based topology optimization. The primary JAX+PETSc realization uses JAX for local energy and constitutive evaluations and PETSc for sparse assembly, Newton globalization, Krylov solvers, and preconditioners. The benchmark suite covers p-Laplace, Ginzburg–Landau, finite-strain hyperelasticity, two- and three-dimensional Mohr–Coulomb plasticity, and topology optimization. Where matched formulations are available, pure JAX and FEniCS provide internal consistency checks. JAX-FEM provides a narrow hyperelastic companion terminal-state comparison, while the Sysala-family slope-stability literature provides reference-model context for the plasticity endpoint observables. The Čermák–Sysala–Valdman MATLAB work supplies elastoplastic finite-element implementation context.

Across these problems, we compare element automatic differentiation, constitutive automatic differentiation, and colored sparse recovery for constructing second-order information, and measure their interaction with PETSc assembly, nonlinear globalization, Krylov solvers, and preconditioners. In the fixed high-order three-dimensional plasticity derivative-route comparison, constitutive AD has the lowest measured wall time while preserving the reported terminal observables. The distributed mechanics examples report MPI timing and solver-policy behavior, and the topology example adds a controlled rank-consistency check under the stated protocol. Together, the evidence shows that derivative construction, sparse assembly, nonlinear globalization, Krylov algebra, and preconditioning must be evaluated as coupled solver-design choices.

1 Introduction

Nonlinear finite-element models in mechanics, scalar PDEs, and design optimization often combine three computational difficulties. The local model may be easiest to write as an energy or constitutive law whose derivatives are tedious to assemble by hand. The global problem then requires robust nonlinear globalization and sparse Newton–Krylov linear algebra. Finally, the implementation must remain comparable across reference formulations, derivative approximations, and distributed-memory solver choices. A useful scientific toolset for this class of problems should therefore expose not only a modeling interface, but also the derivative route, assembly layout, nonlinear solver policy, and preconditioner design.

We develop and evaluate such a scientific toolset for nonlinear finite-element energy minimization and optimization. In the primary JAX+PETSc realization, JAX defines local energies and constitutive laws, while PETSc provides sparse distributed algebra, nonlinear globalization, Krylov methods, and preconditioners. The pure JAX and FEniCS formulations provide internal checks where matched benchmark definitions exist. External comparisons are deliberately scoped: JAX-FEM is used for a narrow hyperelastic companion terminal-state comparison, and the Sysala-family slope-stability literature provides reference-model context for the plasticity endpoint observables. The Čermák–Sysala–Valdman MATLAB work is used as elastoplastic implementation context rather than as a numerical baseline.

1.1 Contributions

Our contribution is a JAX+PETSc scientific toolset and numerical study that treats derivative construction, MPI sparse assembly, nonlinear globalization, and preconditioning as coupled design choices for nonlinear FEM energy minimization. The paper contributes three connected pieces of evidence:

1. **An integrated derivative–assembly path.** We show how local JAX energy or constitutive evaluations can feed PETSc-owned sparse MPI residual, tangent, and Hessian-vector-product computations without changing the benchmark functional being solved.

2. **Controlled derivative-route comparisons.** We compare element automatic differentiation (AD), constitutive AD, and colored sparse recovery inside the same solver realization, so the reported differences reflect derivative construction rather than changes in the underlying discrete model.

3. **Solver-policy evidence across model classes.** We study line-search, trust-region, Krylov, algebraic-multigrid, direct-coarse, and polynomial-multigrid choices on scalar, mechanics, plasticity, and topology benchmarks, with conclusions restricted to the tested benchmark definitions.

Relative to JAX-native differentiable FEM and FEM-adjoint automation, the distinction is the integrated evaluation of local AD routes inside PETSc-owned sparse MPI Newton solvers, with globalization and preconditioner policy varied on a heterogeneous benchmark suite.

We evaluate these components on six benchmark families: p-Laplace, scalar Ginzburg–Landau, hyperelasticity, two- and three-dimensional Mohr–Coulomb-type plasticity, and topology optimization. The scalar problems isolate formulation and derivative agreement. Hyperelasticity adds finite-strain mechanics with a matched JAX-FEM terminal-state companion comparison. The plasticity benchmarks test nonsmooth constitutive response, endpoint-observable comparison, and multigrid behavior. Topology optimization tests a distributed design-and-mechanics loop. The geomechanics context follows the finite-element strength-reduction and 3D slope-stability literature [1–5], but the present plasticity study is limited to final-load endpoint observables rather than path-consistent incremental plastic-history validation.

The surrounding literature covers expressive finite-element automation, differentiable FEM, JAX–solver bridges, mechanics and topology benchmark families, and sparse nonlinear solver infrastructure. The present work combines these roles by treating local differentiation, PETSc-owned sparse MPI assembly, nonlinear globalization, and preconditioner policy in one heterogeneous benchmark suite. Section 2 discusses these source classes by technical role.

The paper is organized as follows. Section 2 deepens the literature context behind this positioning. Section 3 describes the common finite-element energy notation, derivative routes, nonlinear globalization, and sparse derivative recovery. Section 4 explains the primary JAX+PETSc implementation and the narrower pure JAX and FEniCS reference implementations used in selected checks.

Section 5 defines the benchmark suite, with each benchmark separated into literature model and reported discrete problem. Section 6 then reports validation and external-comparison evidence, deliberately separating hyperelastic JAX-FEM agreement from the three-dimensional plasticity endpoint-surrogate comparison. Section 7 reports timing, scaling, derivative-route, and discretization-comparison evidence. Section 8 and Section 9 synthesize the main scope and limitations. Supporting solver-policy diagnostics are collected in Section A.

2 Related Work

Existing systems cover important parts of the nonlinear finite-element calculation: high-level FEM automation, differentiable FEM, solver bridges, mechanics benchmarks, and scalable nonlinear algebra. The gap addressed here is the combined study of local derivative construction, sparse MPI assembly, globalization, and preconditioning on one heterogeneous energy-minimization suite. We therefore organize the relevant literature into four groups: finite-element and PDE-optimization automation, differentiable FEM and JAX–solver bridges, mechanics and topology benchmark context, and sparse nonlinear solver infrastructure.

Table 1 compares the computational roles needed for this positioning: modeling layer, differentiation scope, second-order information, parallel execution model, and benchmark families closest to the present suite. Its main purpose is to show which roles are already covered by existing families and which combination is studied here.

Table 1: Selected related computational roles for the software and literature families most relevant to this paper.

Technical family	Documented role	Relation to this work
FEM and adjoint automation [6–10]	High-level variational forms, generated kernels, adjoint-based sensitivities, and PDE-optimization loops.	Provides the reference high-level FEM and optimization context; selected FEniCS runs are scoped comparison formulations.
JAX-native differentiable FEM and PDE solvers [11–14]	Source-specific roles: GPU-oriented differentiable mechanics (JAX-FEM/JAX-CPFEM), implicit differentiation and solver options (AutoPDEx), and Hessian-vector inverse-problem workflows (Xue 2026).	Motivates local differentiable modeling; the present study instead couples local JAX derivatives to PETSc sparse MPI solves and compares derivative routes.
FEM–JAX and JAX–PETSc bridge architectures [15–17]	Source-specific bridges: Firedrake–JAX variational coupling, FEniCSx external operators for constitutive models, and JetSCI-style JAX–PETSc differentiated local discretizations.	Architectural context for the JAX+PETSc realization; this study evaluates derivative construction, globalization, and preconditioning on the stated benchmark suite.
Slope-stability and elastoplastic benchmark lineage [1–5, 18]	Strength-reduction plasticity, nonsmooth return mapping, continuation, and practical 2D/3D elastoplastic implementation.	Defines the plasticity problem class and lineage; the numerical claims remain limited to the implemented endpoint surrogate and reported comparison observables.
Topology-optimization benchmark lineage [19–23]	SIMP compliance minimization, density filtering, compact educational implementations, and modern parallel topology software.	Defines the topology problem context; the numerical claims are the reported design-and-mechanics timing and rank-consistency diagnostics.

The present JAX+PETSc toolset combines these roles by coupling local JAX energy and constitutive derivatives to

PETSc sparse MPI assembly, nonlinear globalization, Krylov methods, and preconditioner policy on the benchmark suite defined in Section 5.

Finite-element and PDE-optimization automation. The original FEniCS project established the high-level finite-element programming model in which weak forms are expressed symbolically and compiled into efficient kernels [6]. The more recent `DOLFINx` preprint keeps that mathematical abstraction while moving to a data-oriented design with explicit support for modern parallelism, custom kernels, and direct access to lower-level data structures [7]. In PDE-constrained optimization, `dolfin-adjoint` showed that adjoints of high-level transient finite-element programs can be derived automatically from the symbolic problem description [8], and `pyadjoint` documents the FEniCS/Firedrake adjoint framework used to automate such computations in practice [9]. The `cashocs` software paper extends this PDE-optimization thread to shape optimization, optimal control, topology optimization via level sets, and MPI-enabled computations [10]. These works are the natural reference point for expressive finite-element and adjoint-centered automation. The present paper uses that context while focusing instead on second-order routes, nonlinear globalization, and sparse distributed solver policy.

Differentiable FEM and JAX–solver bridges. The broad automatic-differentiation (AD) background is surveyed by Baydin et al. [24], while JAX provides a program-transformation environment for higher-order derivatives, vectorization, and just-in-time compilation in scientific computing [25]. Within finite elements, JAX-FEM demonstrates inverse design, nonlinear constitutive modeling, and GPU-oriented differentiable computations in a JAX-native 3D FE solver [11]. Xue's later finite-element differentiable-physics paper is a close methodological reference for second-order information because it derives implicit Hessian-vector products and evaluates Newton–CG use of exact Hessian information on finite-element inverse-problem benchmarks [12]. AutoPDEx broadens the JAX-native PDE picture with nonlinear minimizers, implicit differentiation, and optional PETSc integration [13]; JAX-CPFEM adds differentiable crystal plasticity and GPU execution in a recent mechanics application [14].

Bridge architectures ask a complementary question: how can JAX-style differentiation be combined with established finite-element or sparse-solver environments? The Firedrake–JAX bridge uses tangent-linear and adjoint equations derived from the high-level PDE representation, avoiding differentiation through low-level solver iterations [15]. The `FEniCSx` external-operator work of Latyshev et al. [16] uses algorithmic automatic differentiation for general constitutive models inside `FEniCSx`, including a Mohr–Coulomb example. The JetSCI preprint is especially close architecturally because it uses JAX for local finite-element discretization and PETSc for robust distributed sparse solves in heterogeneous micromechanics [17]. These works motivate the primary JAX+PETSc realization studied here. The distinction is that we compare element AD, constitutive AD, and colored sparse recovery from AD-based Hessian-vector-product (AD-HVP) or finite-difference gradient probes inside one nonlinear energy-minimization suite spanning the benchmark families in Section 5.

Mechanics and topology benchmark context. The plasticity benchmarks use Mohr–Coulomb-type slope-stability problems, so the relevant literature separates continuum plasticity, strength-reduction practice, constitutive integration, and endpoint comparisons. The Davis-style reduction context used in this paper traces to the plasticity treatment of Davis [26] and to later finite-element strength-reduction studies such as Tschuchnigg et al. [1]. For constitutive integration, the incremental reference frame is return mapping with consistent tangents [27, 28]. The Sysala slope-stability line provides Mohr–Coulomb source-family context, variational and optimization viewpoints for shear strength reduction [2–4], and a 3D continuation/iterative-solver reference line [5]. The Čermák–Sysala–Valdman MATLAB implementation [18] documents practical 2D/3D elastoplastic finite-element algorithms for this problem class. We use these works to position the benchmark family; the numerical claims remain limited to the implemented endpoint surrogate and the reported comparison observables.

For topology optimization, the classical SIMP and educational references supply the compliance, filtering, and continuation ideas used throughout the field [19–22]. A relevant modern software reference is `FEniTop`, which documents a 2D/3D `FEniCSx` topology-optimization implementation with parallel support [23]. It does not duplicate the reduced design objective used here, but it contextualizes compact parallel topology optimization in the modern `FEniCSx` ecosystem.

Sparse derivative recovery and scalable nonlinear infrastructure. When exact second-order information is too expensive, sparse derivative recovery via graph coloring remains the classical alternative. Curtis, Powell, and Reid gave the foundational sparse-Jacobian viewpoint [29], while Coleman and Moré extended the approach to sparse Jacobian and Hessian estimation with graph-coloring structure [30, 31]. PETSc supplies the distributed vectors, matrices, Krylov methods, nonlinear solvers, and multigrid infrastructure needed to turn those derivative objects into scalable distributed algorithms [32, 33]. In the present toolset, PETSc ownership layout, Krylov policy, nonlinear globalization, and preconditioner design are part of the numerical method.

Against that background, the present toolset uses JAX for local element and constitutive differentiation, while PETSc owns sparse distributed algebra and nonlinear solver policy for the benchmark suite in Section 5.

3 Methodology

We use a common finite-element notation across all benchmark families. The continuous domain is $\Omega \subset \mathbb{R}^d$, and $V_{h,0} = \mathrm{span}\{\varphi_i\}_{i=1}^n$ is the conforming space of free basis functions after homogeneous essential constraints have been applied. Let θ collect benchmark parameters such as load, material, continuation, strength-reduction, and lift data. Nonhomogeneous essential data are represented by an affine lift $\bar{u}_h(\theta)$, with $\bar{u}_h = 0$ in the homogeneous cases. For vector-valued mechanics problems, $V_{h,0}$ denotes the constrained vector space, basis functions may be vector-valued, and n counts free scalar degrees of freedom. A free coefficient vector $x \in \mathbb{R}^n$ defines the finite-element state

$$u_h(x;\theta) = \bar{u}_h(\theta) + \sum_{i=1}^{n} x_i \varphi_i.$$

When no confusion is possible, we identify u_h with its coefficient vector in algebraic formulas. On element e, $x_e = R_e x$ denotes the local coefficient vector selected by the element restriction R_e, while $u_e(x;\theta)$ denotes the corresponding local element state including any contribution from the lift.

The benchmark families use two scalar functional roles. A load-step potential or endpoint surrogate is written on a mesh $\mathcal{T}_h = \{e\}_{e=1}^{n_{\mathrm{el}}}$ as

$$\Pi_h(x;\theta) = \sum_{e=1}^{n_{\mathrm{el}}} \Phi_e(x_e;\theta_e) - f_{\mathrm{ext}}^\top x, \tag{1}$$

where θ_e is the element-local restriction of those data and f_{ext} is the assembled load vector on the free degrees of freedom. The expression is the reduced algebraic potential in the free variables; additive constants depending only on the affine lift or prescribed loads are omitted because they do not affect the stationarity equations. A reduced optimization objective with design variable z is written schematically as

$$\mathcal{J}_h(z) = \mathcal{R}_h(z) + \mathcal{S}_h(z, x(z); \theta(z)), \tag{2}$$

where z is a design variable, $x(z)$ is the associated state determined by the benchmark state equation or stationarity condition, and $\mathcal{R}_h$ contains regularization or design penalties. The term $\mathcal{S}_h$ denotes the state-dependent scalar part; in topology optimization it is the frozen compliance term defined in section 5.6, not the load-step potential (1). Thus (1) covers the scalar PDE, mechanics, and endpoint-surrogate solves, while (2) records the reduced-objective role. When an algorithm is stated independently of a benchmark family, we denote its merit functional by $\mathcal{F}$. Hyperelasticity is the only family in which $J = \det F$ denotes the deformation Jacobian, so we avoid the bare symbol J for generic objectives elsewhere in the paper.

In (1), Φ_e is the integrated scalar contribution of element e. In quadrature-based mechanics examples, $\Phi_e(x_e;\theta_e) = \sum_q w_q \mathcal{W}_q(\varepsilon_q;\theta_q)$, where $\varepsilon_q := B_q u_e(x;\theta)$ is the local strain or gradient argument, q indexes quadrature points, w_q includes the quadrature weight and element-map Jacobian factor, B_q is the corresponding element operator, θ_q is the quadrature-point restriction of the element data, and $\mathcal{W}_q$ is the pointwise energy density or branch potential. Within each benchmark family, all derivative routes and solver-realization comparisons use the same quadrature points and weights for the stated discrete functional; the comparisons therefore change the differentiation, assembly, or solver path, not the quadrature definition of the energy. For finite-element labels, $P_k(L_\ell)$ denotes degree-k Lagrange elements on mesh level L_ℓ; problem-specific sections define the mesh hierarchy behind those levels when it matters for interpretation.

3.1 Finite-element and derivative structure

The main design choice in the toolset is that problem definitions should remain energy-centered whenever possible. For branch potentials, all second-order objects below are active-branch derivatives away from switching sets and, for principal-value formulas, away from repeated-principal-value degeneracies. We use three derivative routes:

1. **element AD:** differentiate the full scalar element contribution $\Phi_e(x_e;\theta_e)$ with respect to the element free-DOF vector;
2. **constitutive AD:** differentiate a quadrature-point energy density $\mathcal{W}_q(\varepsilon_q, m_q)$ with respect to ε_q, holding the active branch together with local material and history data $m_q \subset \theta_q$ fixed, then assemble $r_e = \sum_q w_q B_q^\top \sigma_q$ and $K_e = \sum_q w_q B_q^\top C_q B_q$, where $\sigma_q := \partial_\varepsilon \mathcal{W}_q(\varepsilon_q, m_q)$ and $C_q := \partial^2_{\varepsilon\varepsilon} \mathcal{W}_q(\varepsilon_q, m_q)$;
3. **colored sparse recovery:** recover sparse Hessian information through directionally probed gradients or Hessian-vector products (HVPs) and graph coloring.

Element AD is the most literal route and is often the cleanest mathematically. Constitutive AD preserves the same defining scalar-energy expression but changes the differentiation boundary from the full element vector to the local strain state. This is especially effective for high-order plasticity, where the full element Hessian is often much more expensive than the 6×6 constitutive tangent. We use *colored sparse recovery* for the graph-colored second-order route and use *colored SFD* as its compact label. The reported colored-SFD configurations use AD-HVP probes; finite-difference gradient probes are the classical sparse finite-difference variant.

3.2 Newton–Krylov globalization

The nonlinear solves are based on Newton or quasi-Newton updates, but they are not all globalized in the same way. Scalar benchmarks primarily use line-search Newton. Hyperelasticity uses a trust-region method with optional post-step line search. Some benchmark configurations use a hybrid trust-region/line-search algorithm in which a reduced trust-region model provides the direction, while a bounded line search chooses the accepted step. The line-search and trust-region ingredients follow standard globalization ideas [34, Chs. 2–4]; see also Conn et al. [35]. Algorithm 1 is a schematic solver-policy template; the reported runs use the stopping tolerances, caps, globalization choices, and linear-solver/preconditioner choices stated in the benchmark and results tables.

Algorithm 1 Hybrid trust-region / line-search Newton

Given x_0, trust radius Δ_0, tolerances, and evaluation routines for energy, gradient, Hessian assembly or application, and linear solves
for $k = 0, 1, \ldots$ **do**
    Assemble $g_k = \nabla \mathcal{F}(x_k)$
    if convergence criteria are satisfied **then**
        terminate
    end if
    Form or define the Hessian operator $H_k = \nabla^2 \mathcal{F}(x_k)$
    Approximately solve for a Newton-like direction p_k^{N} from $H_k p_k^{\mathrm{N}} = -g_k$
    if trust region is enabled **then**
        Build a small reduced model in the span of Newton and gradient directions
        Solve the reduced trust subproblem with radius Δ_k
        Optionally compare against a pure gradient trust step
        Set d_k to the selected reduced-model direction
        Restrict the line-search interval so the trial step stays in the trust ball
    else
        Set $d_k = p_k^{\mathrm{N}}$
    end if
    Select a trial step $s_k = \alpha_k d_k$ with the stated line-search rule
    Compute $\mathrm{ared}_k = \mathcal{F}(x_k) - \mathcal{F}(x_k + s_k)$ and $\mathrm{pred}_k = -g_k^\top s_k - \frac{1}{2} s_k^\top H_k s_k$
    if trust region is enabled **then**
        accept/reject using $\rho_k = \mathrm{ared}_k/\mathrm{pred}_k$ when $\mathrm{pred}_k > 0$, otherwise reject as model failure; update Δ_k
    else
        accept according to the stated merit-decrease rule
    end if
    if the trial step is accepted **then**
        Set $x_{k+1} = x_k + s_k$
    else
        Retry with the updated globalization state or report a capped/failed solve
    end if
end for

Here ared_k is the achieved decrease in the merit functional and pred_k is the decrease predicted by the local quadratic or reduced trust-region model. A nonpositive predicted decrease is treated as model failure for the trust-region acceptance test. The admissible line-search families are Armijo backtracking, bounded merit search, and residual-bisection safeguards. Benchmark tables instantiate the finite set of remaining policy parameters, including accept/reject thresholds, radius updates, caps, and fallback rules, wherever they affect interpretation. Reported PETSc linear solves use the relative KSP residual tolerance summarized in Table 15 or stated with the corresponding run when the policy varies. Failed linear solves and non-descent directions are handled by the stated globalization safeguard or reported as capped solver behavior.

For the converged bottom-clamped three-dimensional plasticity scaling configuration reported in this paper, the chosen policy is simpler: Newton with Armijo backtracking and a same-mesh polynomial-multigrid $P_4 \to P_2 \to P_1$ hierarchy for the flexible-GMRES linear solve [34, Sec. 3.1]. The reported $\lambda_{\mathrm{sr}} = 1.55$ scaling profile uses a redundant LU/MUMPS coarse solve and PETSc relative KSP tolerance 1×10^{-2}; the auxiliary $\lambda_{\mathrm{sr}} = 1.0$ timing runs use the Hypre-coarse PMG profile. Here λ_{sr} is the plasticity strength-reduction factor defined in section 5.5, PMG denotes same-mesh polynomial multigrid, and KSP denotes PETSc's Krylov-solver interface and residual tolerance.

Algorithm 2 Armijo line search used in Newton solves

```
Given state x, direction p, initial step α_0, backtracking parameters β ∈ (0,1), c_1 ∈ (0,1), and maximum count m_max
Evaluate F(x) and g = ∇F(x)
if g^T p ≥ 0 then
    report direction failure to the calling solver or use the stated fallback
    return
end if
for m = 0, 1, ..., m_max do
    α ← α_0 β^m
    x_trial ← x + αp
    if F(x_trial) ≤ F(x) + c_1 α g^T p then
        accept α
        return
    end if
end for
declare line-search failure or use a problem-specific safeguarded steepest-descent fallback when that policy is enabled
```

The numerical tables use a small common reporting vocabulary, summarized in Table 2. These terms are part of the experimental protocol: a completed endpoint solve, a fixed-work diagnostic, and a capped run do not support the same type of claim.

Table 2: Solver-status and timing vocabulary used in the numerical tables.

Reported term	Definition in this paper	Interpretation
completed	All configured stopping tests for the configuration are met before the nonlinear cap or wall-time cap.	Endpoint energy and observables are terminal values for the stated benchmark.
timeout / iteration cap	The wall-time limit or maximum nonlinear-iteration count is reached before the stated stopping tests are met.	Retained as solver-behavior evidence, not as validation evidence.
fixed work	A prescribed amount of work is run, such as one Newton step, one linearization, or a fixed continuation schedule.	Used for cost, memory, or policy diagnostics rather than endpoint convergence claims.
wall time	Elapsed end-to-end time for the stated run or run segment, including setup when the reporting path includes it.	Comparable only for configurations that share the same benchmark contract and timing scope.
solve time / solver total	Timer internal to the nonlinear or PETSc solve; setup, state export, and postprocessing may be outside this timer.	Used for within-table solver comparisons, not as a universal replacement for wall time.
relative correction	Norm of the accepted Newton correction relative to the current iterate scale; paired gradient checks use the absolute or relative target stated with the result.	Defines the stopping metric in Plasticity3D rows that report correction or gradient targets.

3.3 Colored sparse recovery

Sparse derivative recovery is built around the observation that if columns that do not interfere in the sparsity pattern are perturbed together, one gradient or HVP evaluation can recover multiple Hessian columns, equivalently multiple columns of the Jacobian of the gradient, [29, 31]. The reported colored sparse-recovery path uses distance-2 coloring on the Hessian sparsity graph and AD-HVP probes; the same recovery formula also describes the classical finite-difference gradient-probe variant. This structure is especially effective when combined with overlap-domain assembly and owned-plus-ghost PETSc layouts.

Let $H := \nabla^2\mathcal{F}(x)$ denote the Hessian of the merit functional at the current state, and let $\mathcal{P}(H)$ be its sparsity pattern. A distance-2 coloring partitions the column indices into groups $\mathcal{C}_c$ such that, for each row i, at most one column $j \in \mathcal{C}_c$ satisfies $(i,j) \in \mathcal{P}(H)$. For the color seed $v_c = \sum_{j\in\mathcal{C}_c} e_j$, one probe produces

$$y_c = \begin{cases} \big(\nabla\mathcal{F}(x+\delta v_c) - \nabla\mathcal{F}(x)\big)/\delta, & \text{finite-difference gradient probe,} \\ Hv_c, & \text{AD Hessian-vector probe.} \end{cases} \quad (3)$$

In the finite-difference branch, $\delta > 0$ is the chosen gradient-probe step size. The structural separation then recovers $H_{ij} = (y_c)_i$ for $j \in \mathcal{C}_c$ and $(i,j) \in \mathcal{P}(H)$ in the AD-HVP branch. In the finite-difference branch it gives $H_{ij} = (y_c)_i + O(\delta)$ under local smoothness. The reported JAX+PETSc colored sparse-recovery cases use the AD Hessian-vector probe mode, so no finite-difference perturbation parameter is introduced in those runs; the finite-difference formula states the classical recovery variant. After recovery, each rank scatters only owned sparse entries to the PETSc matrix, with overlap data used only for local probe evaluation.

Algorithm 3 Distributed colored Hessian recovery

```
Construct the owned part of the Hessian sparsity pattern
Compute a distance-2 coloring of the Hessian pattern
for each color group c do
    Build the combined probe vector v_c
    Evaluate one gradient difference or HVP for v_c
    Scatter recovered entries into owned sparse storage
end for
Assemble the owned rows of the distributed sparse matrix
```

3.4 Constitutive autodiff for 3D plasticity

The main Plasticity3D result in this paper uses constitutive AD. Rather than applying second-order AD to the full P_4 element vector, we define one scalar Mohr–Coulomb branch potential $\mathcal{W}_q(\varepsilon_q, m_q)$ per quadrature point, where m_q collects the active branch together with the local material and history data selected from θ_q. We differentiate this scalar potential with respect to the six engineering-strain components and assemble the element tangent from the resulting stress and constitutive tangent. This keeps the scalar-energy definition intact while reducing the cost of branchwise second-order information on high-order tetrahedra. The operator B_q, stress vector σ_q, and tangent C_q use the same Voigt engineering-shear convention, so σ_q is dual to the chosen strain vector.

Algorithm 4 Constitutive AD assembly for high-order plasticity

for each local element e **do**
    for each quadrature point q **do**
        Evaluate strain $\varepsilon_q = B_q u_e(x;\theta)$
        Evaluate scalar energy density $\mathcal{W}_q(\varepsilon_q, m_q)$
        Obtain stress $\sigma_q = \partial_\varepsilon \mathcal{W}_q(\varepsilon_q, m_q)$
        Obtain tangent $C_q = \partial^2_{\varepsilon\varepsilon} \mathcal{W}_q(\varepsilon_q, m_q)$
    end for
    Assemble $r_e = \sum_q w_q B_q^\top \sigma_q$
    Assemble $K_e = \sum_q w_q B_q^\top C_q B_q$
end for

The key methodological point is that automatic differentiation is applied directly to the scalar branch potential defined by the benchmark. The constitutive tangent is therefore obtained from that same scalar potential rather than replaced by a separate hand-derived formula. Away from branch switching sets and repeated-principal-value degeneracies, this yields branchwise exact derivatives of the surrogate; on those nonsmooth sets the resulting tangents are branchwise surrogate derivatives rather than globally smooth continuum derivatives.

4 Implementation

The computational realization starts from a self-contained JAX+PETSc pipeline. A benchmark supplies a local energy density, branch potential, or constitutive law together with finite-element restrictions. JAX evaluates the local residual, tangent, or Hessian-vector-product contribution on each element or quadrature point. The distributed layer scatters owned coefficients to ghosted element states, accumulates owned-row sparse residuals and tangents, and gives PETSc the resulting nonlinear residual, matrix, or matrix-free action. PETSc then controls globalization, Krylov iteration, preconditioning, and convergence reporting. The same pipeline is used before narrower reference formulations are introduced for selected benchmark checks.

Primary JAX+PETSc realization. This is the central solver realization of the paper. It combines JAX-based local physics with PETSc distributed vectors, matrices, Krylov solvers, and preconditioners. Problem-specific local evaluations stay in JAX; sparse distributed assembly, nonlinear globalization, and linear algebra stay in PETSc.

Pure JAX reference implementations. These are compact differentiable realizations of selected benchmark energies. They are useful for serial formulation checks and AD diagnostics where a matched serial formulation exists, and for compact design demonstrations where the benchmark role is narrower. They are not intended to be the primary distributed route for the largest distributed benchmarks. Their main scientific role in this paper is to show how selected energy formulations can also be expressed without sparse distributed infrastructure.

FEniCS reference implementations. The FEniCS implementations remain valuable on families where they exist, especially p-Laplace, Ginzburg–Landau, and hyperelasticity. They provide a trusted high-level FEM reference and help distinguish modeling issues from solver-policy issues. In this paper they are used mainly as scoped reference comparisons rather than as the primary distributed realization.

4.1 Autodiff modes

The JAX+PETSc realization exposes several second-order modes. The precise available choices depend on the benchmark family, but the common logic is:

- **element AD**: local Hessians obtained by differentiating the full element energy on the active smooth branch;
- **constitutive AD**: branchwise local constitutive tangents assembled from quadrature-point energy densities;
- **colored sparse recovery**: structured sparse recovery from AD-HVP or finite-difference gradient probes when full element Hessians are not the appropriate cost tradeoff. We refer to this sparse finite-difference-style recovery route as colored SFD.

This split is scientifically important. It lets us compare AD-derived local second-order information against carefully structured sparse recovery routes without changing the nonlinear solver or sparse assembly path. Only the finite-difference probe variant is an approximation to the directional derivative. In other words, the paper compares derivative strategies inside one common distributed solver construction, not only across reference implementations.

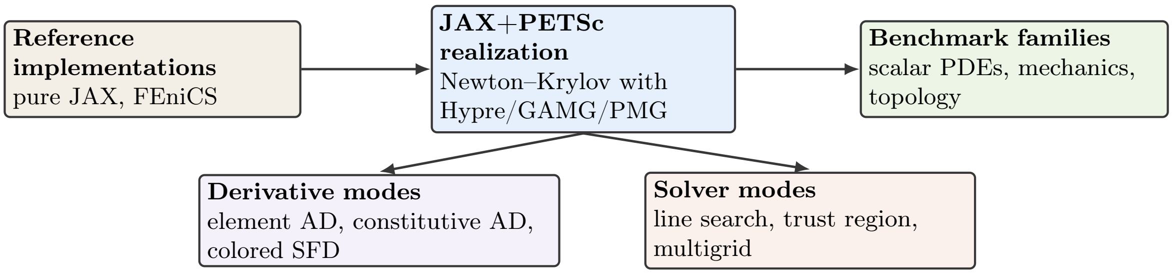


Figure 1: Computational structure used in the study. The JAX+PETSc formulation is the distributed realization; pure JAX and FEniCS provide reference formulations where matched benchmark definitions are available.

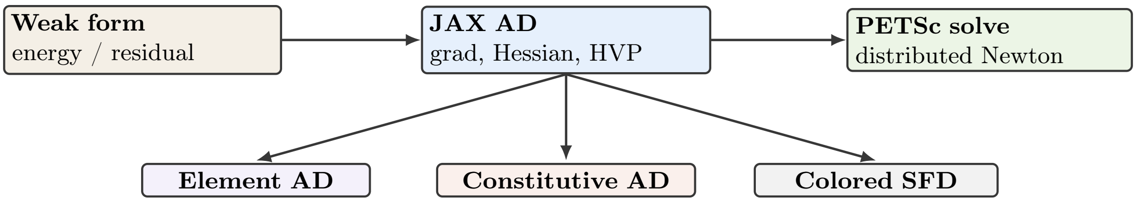


Figure 2: Derivative pathways from scalar energies and quadrature-point potentials to distributed PETSc nonlinear solves.

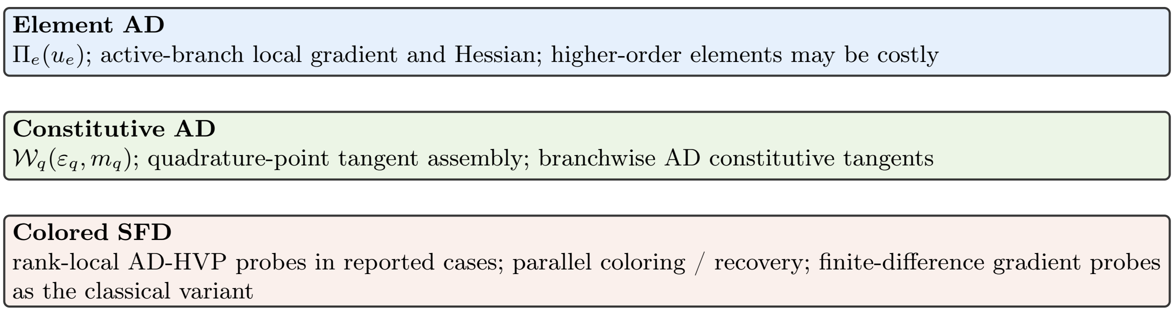


Figure 3: Second-order derivative modes exposed in the JAX+PETSc realization. The methodological choice is that the scalar energy or energy-density expression remains the defining mathematical object even when the tangent route changes from element AD to constitutive AD or colored sparse recovery.

Figures 1 to 3 summarize the computational construction used in the rest of the paper. The primary JAX+PETSc realization is the distributed nonlinear solve; the reference formulations provide matched comparisons; and the derivative modes differ only in where the second-order information is constructed before PETSc receives sparse algebraic objects.

4.2 Linear solvers and preconditioners

The benchmark families use different Krylov and preconditioner combinations because the Hessian and tangent systems have different definiteness, block structure, and coarse-level behavior. Here CG denotes conjugate gradients, GMRES generalized minimal residual, FGMRES flexible GMRES, GAMG PETSc's geometric-algebraic multigrid preconditioner, Hypre an algebraic multigrid/preconditioning backend, MUMPS a multifrontal sparse direct solver, and PMG polynomial multigrid:

- **p-Laplace:** the scalar positive-definite structure makes Newton–CG with algebraic multigrid a natural baseline for comparing exact element Hessians and colored sparse-recovery Hessians.

- **Ginzburg–Landau:** the double-well energy produces more indefinite linearizations, so GMRES with algebraic multigrid is used to separate globalization behavior from symmetric-positive-definite assumptions.
- **Hyperelasticity:** large-deformation mechanics requires globalization around the logarithmic energy domain, while the linearized systems motivate GAMG and same-hierarchy PMG variants with robust coarse solves.
- **Plasticity:** nonsmooth branch structure and high-order element spaces motivate flexible Krylov methods and same-mesh polynomial multigrid hierarchies, with direct or algebraic coarse solves used to expose the cost of the active constitutive tangent.
- **Topology:** the mechanics subproblem is solved with flexible Krylov iteration and GAMG inside an outer distributed design update, so the solver evidence reflects the coupled state-design workflow.

The result sections then instantiate these solver choices in concrete benchmark rows. Some rows deliberately hold the preconditioning operator fixed along the nonlinear path or vary only the coarse solve. These diagnostics are included to separate derivative-route cost, preconditioner construction, and outer globalization effects.

4.3 Distributed assembly and coloring

The JAX+PETSc implementation uses PETSc-owned ranges, local-plus-ghost layouts, and owned-row sparse assembly. Let I_r be the free-DOF indices owned by MPI rank r, with the sets $\{I_r\}$ forming a disjoint partition of the global free indices. Let G_r be the ghost indices needed by elements that touch I_r, and let $\widetilde{I}_r = I_r \cup G_r$. Before energy, gradient, Hessian, or HVP work, owned coefficients are scattered to $\widetilde{I}_r$; the JAX evaluations then compute element and quadrature contributions rank-locally on this overlap state.

Matrix entries are inserted only for rows in I_r, with values on G_r used to evaluate local stencils but not to duplicate ownership. Scalar energy values reported in the results are the solver-reported global reductions for the corresponding configuration; the ownership contract here concerns derivative and matrix assembly. This communication order matters for both direct second-order assembly and colored sparse recovery. Exact element or constitutive assembly can write directly into local sparse buffers. Colored sparse recovery requires an additional graph-coloring step, but once the coloring is known the AD-HVP or finite-difference probe evaluations use the same ghost fill on $\widetilde{I}_r$ and still produce owned-row sparse insertions.

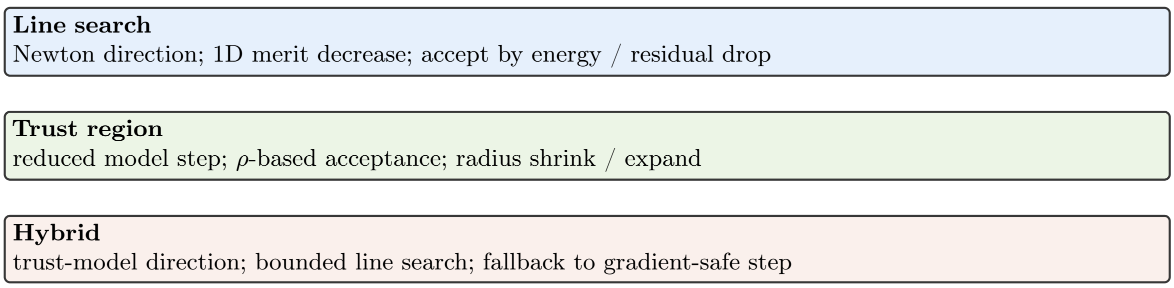


Figure 4: Globalization choices used across the benchmark suite, from pure line search to trust-region and hybrid strategies.

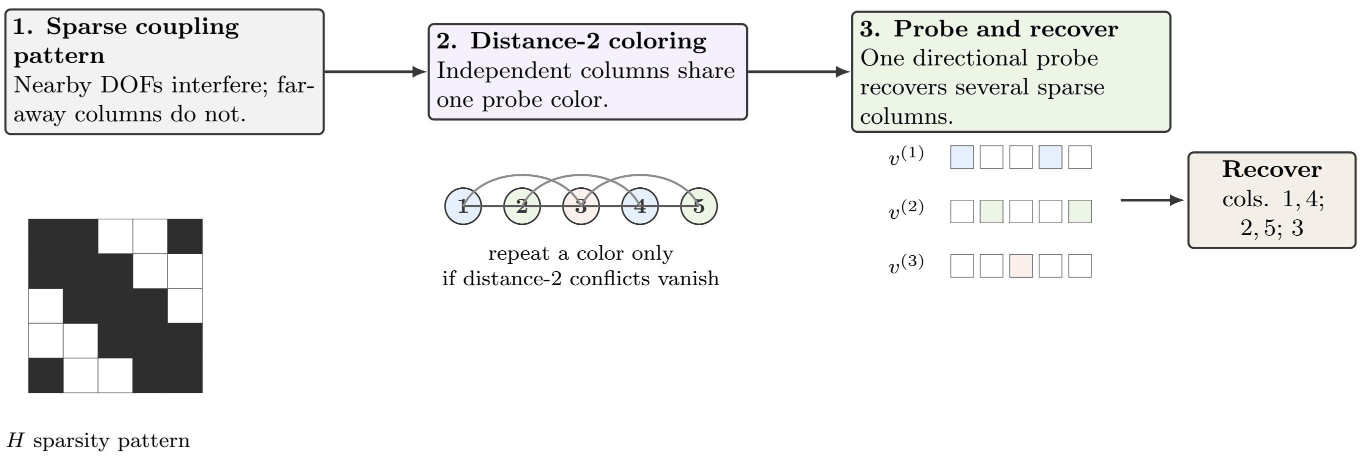


Figure 5: Schematic view of the colored sparse-recovery logic used by the colored Hessian-recovery path: a sparsity pattern, a distance-2 coloring, and the corresponding grouped probes used for recovery.

Figures 4 and 5 make two solver-policy choices explicit. Globalization is varied independently of the element physics where the benchmark permits it, while colored sparse recovery uses graph structure to batch directional Hessian probes without changing the nonlinear problem definition.

Table 3: Solver and derivative components by benchmark family. "Solver and derivative components" summarizes the derivative routes, nonlinear policies, or coupled solver features that are scientifically relevant in each family.

Family	FEniCS	pure JAX	JAX+PETSc	Solver and derivative roles
p-Laplace	yes	yes	yes	element AD and colored sparse recovery; Newton–CG with Hypre
Ginzburg–Landau	yes	no	yes	element AD and colored sparse recovery; Armijo Newton with GMRES–Hypre
Hyperelasticity	yes	yes	yes	element AD in the primary route, scoped colored sparse-recovery comparison, and trust-region/GAMG or PMG diagnostics
Plasticity2D	no	no	yes	endpoint branch-potential derivatives and continuation diagnostics with a same-mesh PMG hierarchy
Plasticity3D	no	no	yes	constitutive AD, element AD, and colored sparse-recovery diagnostics; FGMRES with same-mesh PMG and Hypre or LU/MUMPS coarse profiles
Topology optimization	no	yes	yes	distributed design updates with PETSc mechanics and GAMG-preconditioned FGMRES

Table 3 summarizes the solver components used in the benchmark suite. The most important structural point is that the JAX+PETSc column is the distributed algebraic realization, even when a family also has FEniCS or pure JAX reference formulations. The table is intentionally a component map rather than a full parameter schedule; run-specific tolerances, caps, timing scopes, and coarse-solver variants are reported with the numerical evidence they qualify.

5 Benchmark Problems

The benchmark suite is intentionally heterogeneous. The central claim of the paper is not that one solver policy dominates every nonlinear PDE family, but that one differentiable sparse-solver toolset can cover scalar elliptic problems, phase-field-type energies, finite-strain mechanics, elastoplastic slope stability, and reduced topology optimization through a common energy and reduced-objective viewpoint.

Table 4: Benchmark families, tested scopes, solver policies, and principal comparison evidence. The later numerical section revisits the same families from the timing and scaling viewpoint.

Family	Tested scope	Solve policy	Comparison evidence	Main difficulty
p-Laplace	L_5 serial agreement; L_9 distributed scaling	Newton + line search	FEniCS and JAX+PETSc on the distributed case; pure JAX in the serial agreement case	nonlinear elliptic solve with exact sparse Hessians
Ginzburg–Landau	L_5 agreement; L_9 distributed scaling	Newton + line search	FEniCS and JAX+PETSc comparison	indefinite local curvature from the double well
Hyperelasticity	L_1 serial agreement; L_4, 24-step distributed scaling	trust-region solve	FEniCS and JAX+PETSc on the distributed suite; pure JAX only as a serial formulation check	nonconvex large-deformation mechanics
Plasticity2D	$P_4(L_5)$–$P_4(L_7)$	endpoint solve and capped fixed work	JAX+PETSc endpoint and solver-policy evidence	same-mesh PMG and nonlinear tail behavior
Plasticity3D	$P_1/P_2/P_4$	endpoint, scaling, and PMG-policy studies	constitutive AD, closed-form tangent assembly, and PMG-policy evidence	heterogeneous 3D Mohr–Coulomb with constitutive AD
Topology	192×96 serial demonstration; 768×384 parallel benchmark	adaptive continuation	pure JAX on the serial demonstration; JAX+PETSc on the fine-grid rank-varied adaptive timing run and controlled rank-consistency check	distributed design-mechanics coupling

Table 5: Availability of FEniCS and pure JAX references. Reference implementations are used where documented implementations exist for the same benchmark family; absent cells are left absent rather than filled with speculative baselines.

Family	FEniCS	pure JAX	Notes
p-Laplace	yes	yes	All three implementations exist; pure JAX is used in the serial parity case.
Ginzburg–Landau	yes	no	FEniCS and JAX+PETSc form the reported comparison.
Hyperelasticity	yes	yes	pure JAX is a serial formulation check only.
Plasticity2D	no	no	The reported realization is JAX+PETSc only.
Plasticity3D	no	no	Closed-form constitutive assembly is used only as a supporting comparison route.
Topology	no	yes	Parallel fine-grid realization is JAX+PETSc; pure JAX remains a compact serial design demonstration.

Tables 4 and 5 define the comparison scope before the individual models are introduced. The scalar problems and hyperelasticity have FEniCS reference formulations, while pure JAX is used only where a compact serial formulation or demonstration is available.

The two plasticity families are primarily JAX+PETSc studies. The two-dimensional case supplies supporting continuation diagnostics, and the three-dimensional case uses the Čermák–Sysala–Valdman MATLAB elastoplastic implementation as implementation context. The Sysala-family slope-stability literature supplies the reference-model setting. Neither source class is used as a path-history validation baseline.

Topology optimization uses a parallel JAX+PETSc realization and a compact serial pure JAX design demonstration, but the agreement evidence for that family is the JAX+PETSc rank-consistency check. Thus some entries below are validation comparisons, whereas others are timing or solver-policy diagnostics under a stated benchmark definition. Table 6 records the compact level notation used in the model definitions and numerical tables; the labels identify discrete problem size and mesh construction, not separate physical models.

Family	Level label	Construction	Representative sizes used below
p-Laplace	L_ℓ	Uniform triangular hierarchy on the L-shaped domain.	L_9: 788,481 nodes, 1,572,864 triangles, 784,385 free scalar DOFs; L_{10}: 3,149,825 nodes, 6,291,456 triangles, 3,141,633 free scalar DOFs.
Ginzburg–Landau	L_ℓ	Uniform triangular hierarchy on the square domain.	L_9: 1,050,625 nodes, 2,097,152 triangles, 1,046,529 free scalar DOFs; L_{10}: 4,198,401 nodes, 8,388,608 triangles, 4,190,209 free scalar DOFs.
Hyperelasticity	L_ℓ	Tetrahedral cantilever hierarchy with vector displacement DOFs.	L_4: 185,249 nodes, 983,040 tetrahedra, 554,013 free displacement DOFs; L_5: 1,395,009 nodes, 7,864,320 tetrahedra, 4,178,493 free displacement DOFs.
Plasticity2D	$P_4(L_\ell)$	Degree-four same-mesh Lagrange space on the homogeneous slope hierarchy.	$P_4(L_5)$–$P_4(L_7)$ have 332,960, 1,331,520, and 5,325,440 free displacement DOFs on 20,800, 83,200, and 332,800 triangles.
Plasticity3D	$P_k(L_\ell)$	Degree-k same-mesh Lagrange space on the base heterogeneous tetrahedral slope and one, two, or three uniform refinements.	The P_1 levels L_1–L_4 have 10,526, 79,024, 610,964, and 4,801,816 free displacement DOFs; P_2 and P_4 use the same macro tetrahedra with elevated element degree.

Table 6: Meaning of the mesh-level labels used in the benchmark and results tables. The counts identify the discrete problem sizes behind the compact labels; they are not additional numerical results.

5.1 p-Laplace

The literature model is the standard stationary p-Laplace Dirichlet problem and its associated energy formulation [36, Ch. 2]. The unknown is a scalar field $u : \Omega \to \mathbb{R}$ and the continuous energy is

$$\mathcal{E}(u) = \int_\Omega \frac{1}{p}|\nabla u|^p \,\mathrm{d}x - \int_\Omega f u \,\mathrm{d}x, \tag{4}$$

whose Euler–Lagrange equation is

$$-\nabla \cdot \left(|\nabla u|^{p-2}\nabla u\right) = f \quad \text{in } \Omega, \qquad u = 0 \quad \text{on } \partial\Omega. \tag{5}$$

The weak problem is: find $u \in W_0^{1,p}(\Omega)$ such that

$$\int_\Omega |\nabla u|^{p-2}\nabla u \cdot \nabla v \,\mathrm{d}x = \int_\Omega f v \,\mathrm{d}x \qquad \forall v \in W_0^{1,p}(\Omega). \tag{6}$$

The benchmark used in this paper takes the L-shaped domain $\Omega = (0,2)^2 \setminus [1,2]^2$, exponent $p = 3$, forcing $f = -10$, and a conforming P_1 triangular hierarchy. Here and below, $\mathcal{E}_h$ denotes the finite-element restriction of $\mathcal{E}$ evaluated with the quadrature used for the numerical comparisons. The discrete problem is

$$u_h \in V_h \subset W_0^{1,p}(\Omega): \qquad u_h = \arg\min_{v_h \in V_h} \mathcal{E}_h(v_h), \tag{7}$$

where V_h is the chosen conforming finite-element subspace. This is the cleanest benchmark in the paper for checking exact gradients, exact Hessians, and sparse-recovery alternatives against each other without constitutive or geometric complications.

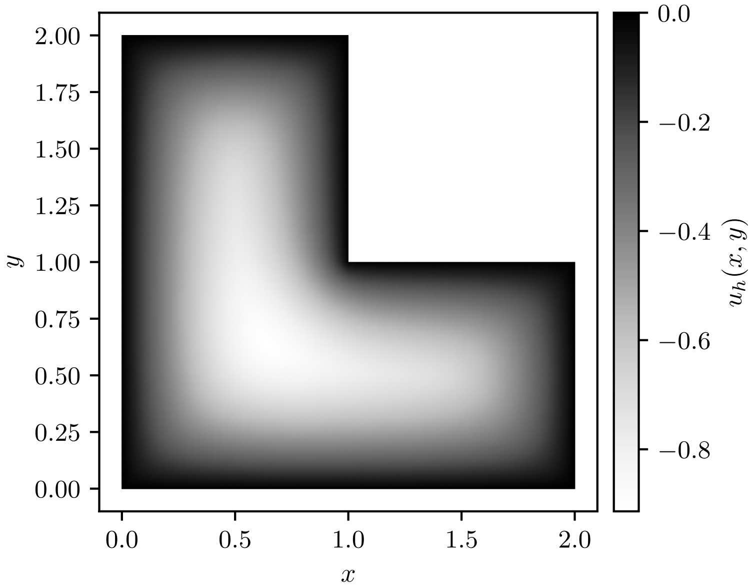


Figure 6: Representative p-Laplace state on the selected hierarchy.

Figure 6 shows the representative state on the L-shaped domain. The visible corner-driven variation is the geometric feature against which the derivative routes are compared in the energy plot and summary table.

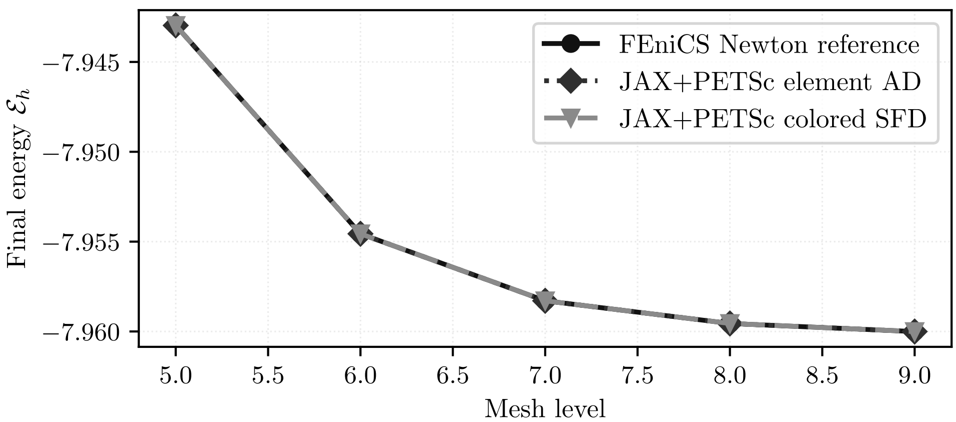


Figure 7: Final discrete energies from (7) across the selected mesh hierarchy. The element-AD and colored sparse-recovery JAX+PETSc paths are energy-consistent with the FEniCS reference at the plotted scale.

Table 7: Representative final discrete energies from (7) on the selected case.

Solver realization	Energy	Newton iters	Krylov iters	Wall time [s]
FEniCS Newton reference	−7.942968719	5	15	0.0782
JAX+PETSc element AD	−7.942968719	6	17	0.0798
JAX+PETSc colored SFD	−7.942968711	6	6	0.194

The within-benchmark picture is therefore one of formulation agreement: figure 7 and table 7 report JAX+PETSc discrete energies within about 1×10^{-8} of the FEniCS Newton reference for (7). Scaling and timing for this family are reported in Section 7.3.

5.2 Ginzburg–Landau

We cite Ginzburg and Landau [37] for the historical origin of Ginzburg–Landau free-energy modeling only. The benchmark used here is a scalar real-valued double-well problem on $\Omega = [-1, 1]^2$ with homogeneous Dirichlet boundary conditions and energy

$$\mathcal{E}(u) = \int_\Omega \frac{\varepsilon}{2}|\nabla u|^2 + \frac{1}{4}(u^2 - 1)^2 \,\mathrm{d}x, \qquad \varepsilon = 10^{-2}. \tag{8}$$

Its strong form is

$$-\varepsilon \Delta u + u(u^2 - 1) = 0 \quad \text{in } \Omega, \qquad u = 0 \quad \text{on } \partial\Omega, \tag{9}$$

and the weak problem is: find $u \in H_0^1(\Omega)$ such that

$$\int_\Omega \varepsilon \nabla u \cdot \nabla v + (u^2 - 1)uv \,\mathrm{d}x = 0 \qquad \forall v \in H_0^1(\Omega). \tag{10}$$

The discrete problem is a conforming P_1 stationary problem on its uniform triangular hierarchy:

$$u_h \in V_h : \qquad D\mathcal{E}_h(u_h)[v_h] = 0 \qquad \forall v_h \in V_h. \tag{11}$$

Relative to p-Laplace, the additional difficulty here is not geometry but indefinite local curvature from the double-well potential in (8). All compared routes use the same initial state

$$u_0(x, y) = \sin\left(\frac{\pi(x-1)}{2}\right) \sin\left(\frac{\pi(y-1)}{2}\right),$$

restricted to the free degrees of freedom, so the nonconvex branch selection is part of the benchmark definition.

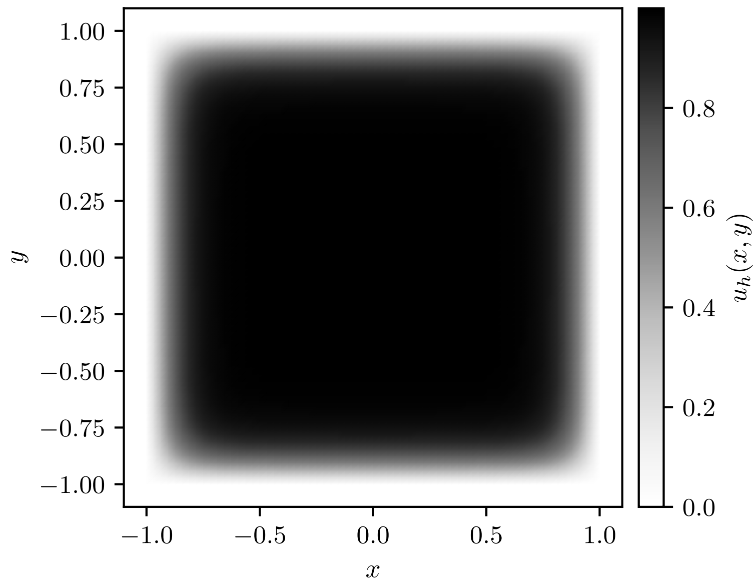


Figure 8: Representative Ginzburg–Landau state on the selected hierarchy.

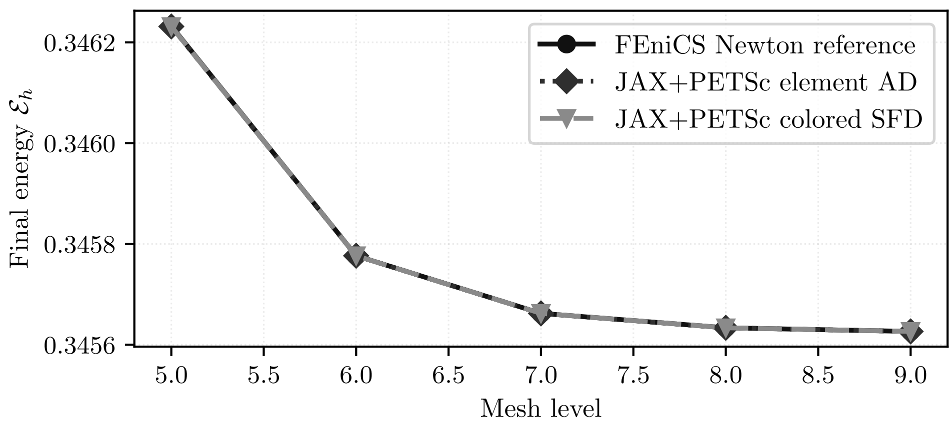


Figure 9: Final discrete energies associated with the stationary problem (11) across mesh levels. The element-AD and colored sparse-recovery JAX+PETSc routes agree with the FEniCS reference within about 1×10^{-6} in final energy.

Table 8: Representative final discrete energies associated with (11) on the selected case.

Solver realization	Energy	Newton iters	Krylov iters	Wall time [s]
FEniCS Newton reference	0.3462323651	8	33	0.113
JAX+PETSc element AD	0.3462314438	8	33	0.0949
JAX+PETSc colored SFD	0.3462314438	8	33	0.247

Figures 8 and 9 and Table 8 show terminal-state consistency from the prescribed initial state. The local picture is again one of formulation consistency, now for a more solver-sensitive nonconvex energy: the FEniCS reference, element AD, and colored sparse recovery all report 8 Newton iterations, 33 Krylov iterations, and final energies within about 1×10^{-6}. Strong scaling and solver-time comparisons for this family are reported in Section 7.4.

5.3 Hyperelasticity

Hyperelasticity is the first benchmark where finite-strain mechanics and nonlinear globalization become central. The literature model is a standard finite-strain equilibrium problem written in terms of a displacement u and deformation $y_u(X) = X + u(X)$. We define the deformation gradient $F(u) = I + \nabla u$, its determinant $J(u) = \det F(u)$, and the compressible Neo-Hookean-type density

$$W(F) := C_1 \left(\mathrm{tr}(F^\top F) - 3 - 2\ln\det F\right) + D_1(\det F - 1)^2.$$

The first Piola stress is $P(F) = \partial W/\partial F$ [38, Chs. 4–6].

The benchmark considered here is a cantilever on $\Omega = [0, 0.4] \times [-0.005, 0.005]^2$ with $C_1 \approx 3.846 \times 10^7$ and $D_1 \approx 8.333 \times 10^7$. On admissible displacements with $J(u) > 0$ almost everywhere, the elastic potential is

$$\Pi(u) = \int_\Omega W(F(u))\,\mathrm{d}x, \tag{12}$$

which is the quantity compared in the benchmark summaries.

The left face is clamped, while the right face carries a prescribed rotating Dirichlet displacement path. Let Γ_L and Γ_R denote the left and right end faces, respectively. For $X = (X_1, X_2, X_3) \in \Gamma_\mathrm{R}$ and $N = 24$ load steps, the prescribed position at step k is

$$\bar{y}_k(X) = \left(X_1, \cos\eta_k\, X_2 + \sin\eta_k\, X_3, -\sin\eta_k\, X_2 + \cos\eta_k\, X_3\right), \qquad \eta_k = \frac{8\pi k}{N}.$$

Thus admissible displacements satisfy $u = 0$ on Γ_L and $X + u = \bar{y}_k(X)$ on Γ_R.

Let $V(\eta_k)$ denote the corresponding affine displacement space, let V_0 denote the homogeneous test space, and let $\Gamma_N = \partial\Omega \setminus (\Gamma_\mathrm{L} \cup \Gamma_\mathrm{R})$. The strong form is

$$-\nabla \cdot P(F(u)) = 0 \quad \text{in } \Omega, \qquad u = 0 \quad \text{on } \Gamma_\mathrm{L}, \qquad X + u = \bar{y}_k(X) \quad \text{on } \Gamma_\mathrm{R}, \qquad P(F(u))n = 0 \quad \text{on } \Gamma_N, \tag{13}$$

where $P = \partial W/\partial F$ is the first Piola stress. Equivalently, the weak form is to find $u \in V(\eta_k)$ such that

$$\int_\Omega P(F(u)) : \nabla v\,\mathrm{d}x = 0 \qquad \forall v \in V_0. \tag{14}$$

Trial states with nonpositive determinant fall outside the logarithmic energy domain and are rejected by the nonlinear globalization; accepted line-search or trust-region steps are therefore evaluated only on states with positive $J(u)$.

For the discrete problem, let $V_h(\eta_k) \subset V(\eta_k)$ be the first-order tetrahedral vector finite-element space, let $V_{h,0} \subset V_0$ be its homogeneous subspace, and let Π_h be the finite-element discretization of (12). We solve the sequence of stationary points

$$u_h^{(k)} \in V_h(\eta_k): \qquad D_u\Pi_h(u_h^{(k)}; \eta_k)[v_h] = 0 \qquad \forall v_h \in V_{h,0}, \qquad k = 1, \ldots, 24, \tag{15}$$

with the boundary data above. This sequence is the basis for the hyperelastic terminal-energy, scaling, and globalization comparisons reported below.

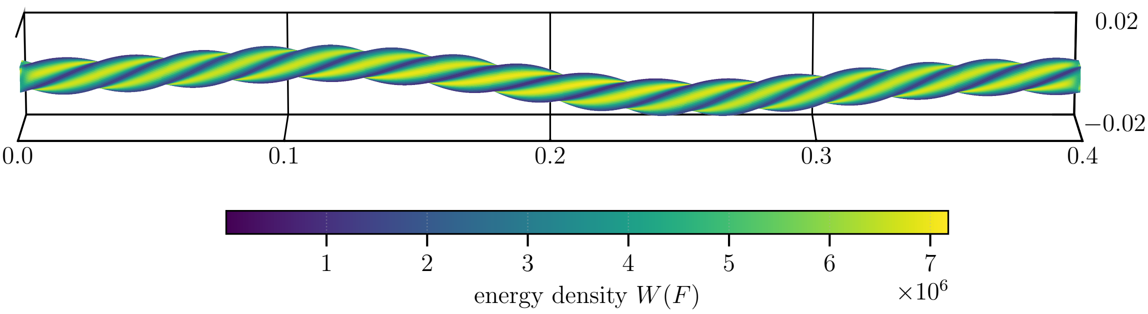


Figure 10: Representative finite-strain hyperelastic state, colored by the density W entering (12).

Table 9: Representative 24-step terminal energies along (15).

Solver realization	Energy	Steps	Krylov iters	Wall time [s]
FEniCS Newton reference	197.775074	24	8722	6.51
JAX+PETSc element AD	197.754582	24	10595	9.36
pure JAX serial formulation check	197.749509	24	2284	41.5

The within-benchmark takeaway is therefore that all three implementations complete the same prescribed 24-step boundary schedule and produce comparable terminal energies. Figures 10 and 11 show the terminal deformed state and energy trend, and Table 9 quantifies the terminal energies and work counts. This evidence does not assess trajectory-level agreement or produce a performance ranking. Globalization and scaling evidence are deferred to Section 7.5. The validation section separately compares a level-1 translation companion case against JAX-FEM under the common postprocessed energy; it is not part of the rotating-load-path evidence.

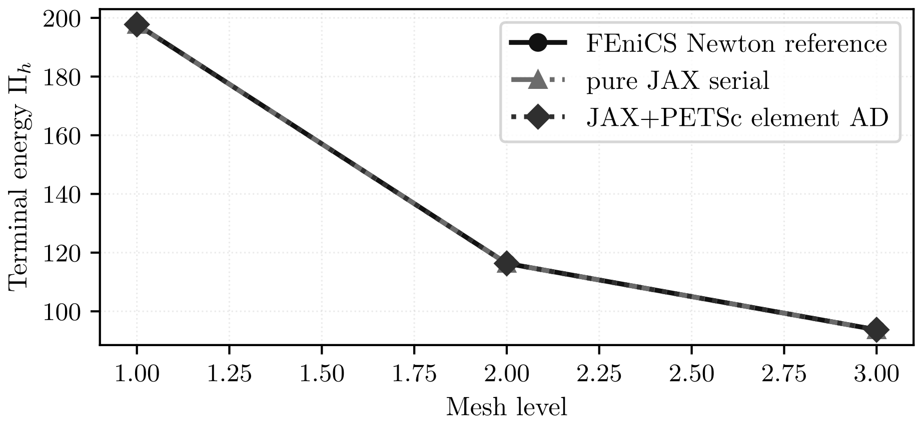


Figure 11: Final discrete energies along the stationary sequence (15) across the selected mesh hierarchy. The comparison records completion of the same 24-step schedule and comparable terminal energies; this figure does not assess trajectory-level agreement.

5.4 Plasticity2D

The literature model is a quasi-static plane-strain Mohr–Coulomb slope stability problem posed as equilibrium together with a constitutive update and history variables [1–3, 26–28]. The unknown is the displacement field $u = (u_x, u_y) : \Omega \to \mathbb{R}^2$ on a homogeneous slope geometry. At increment $n+1$, the continuum equilibrium equations are

$$-\nabla \cdot \sigma^{n+1} = b^{n+1} \quad \text{in } \Omega, \qquad u_x^{n+1} = \bar{u}_x^{n+1} \quad \text{on } \Gamma_{D,x}, \qquad u_y^{n+1} = \bar{u}_y^{n+1} \quad \text{on } \Gamma_{D,y}, \qquad \sigma^{n+1} n = t^{n+1} \quad \text{on } \Gamma_N, \quad (16)$$

where σ^{n+1} and the internal variables are obtained from the elastoplastic update driven by $\varepsilon(u^{n+1})$ and the previous state.

The computational slope has the standard homogeneous benchmark geometry specified by lower bench width 15, slope run 10, upper bench width 15, two vertical height segments of length 10, and slope angle 45°. The bottom edge is fixed in both displacement components, the left and far-right vertical sides are fixed horizontally, and the sloping top is traction-free. The material parameters are $E = 40{,}000$, $\nu = 0.30$, $c_0 = 6.0$, $\phi = 45°$, $\psi = 0°$, and gravity magnitude $\gamma = 20$, so $b = (0, -\gamma)^\top$ in the endpoint solves.

The endpoint studies set the old plastic strain to zero at all quadrature points. Let Γ_{bot} denote the bottom edge and let Γ_{side} denote the constrained vertical side edges. We use $\Gamma_{D,x} = \Gamma_{\text{bot}} \cup \Gamma_{\text{side}}$ and $\Gamma_{D,y} = \Gamma_{\text{bot}}$, so the homogeneous test space V_0 contains functions with $v_x = 0$ on $\Gamma_{D,x}$ and $v_y = 0$ on $\Gamma_{D,y}$. The traction boundary is $\Gamma_N = \partial\Omega \setminus (\Gamma_{\text{bot}} \cup \Gamma_{\text{side}})$. The corresponding weak form is

$$\int_\Omega \sigma^{n+1} : \varepsilon(v)\,\mathrm{d}x = \int_\Omega b^{n+1} \cdot v\,\mathrm{d}x + \int_{\Gamma_N} t^{n+1} \cdot v\,\mathrm{d}s \qquad \forall v \in V_0. \quad (17)$$

The computations reported here are endpoint solves, not path-consistent incremental histories. Accordingly, we first define the local Davis-B reduction and branch potential, and then assemble them into the scalarized discrete surrogate in (19).

The two-dimensional branch formulas use engineering strain $\varepsilon = (\varepsilon_{xx}, \varepsilon_{yy}, \gamma_{xy})^\top$ and the plane-strain elastic matrix

$$C_{2\text{D}} = \begin{bmatrix} \lambda_{\text{ps}} + 2\mu & \lambda_{\text{ps}} & 0 \\ \lambda_{\text{ps}} & \lambda_{\text{ps}} + 2\mu & 0 \\ 0 & 0 & \mu \end{bmatrix}, \qquad \lambda_{\text{ps}} = \frac{E\nu}{(1+\nu)(1-2\nu)}, \qquad \mu = \frac{E}{2(1+\nu)}.$$

Its compliance is $S_{2\text{D}} = C_{2\text{D}}^{-1}$. Principal stresses are ordered as $\sigma_1^{\text{tr}} \geq \sigma_2^{\text{tr}}$.

The benchmark uses a Davis-B-style reduction path, with the Davis citation used as historical lineage and the formula context taken from later strength-reduction descriptions [1, 3, 26], to map the unreduced cohesion and angles to reduced values under strength reduction. Let $\lambda_{\text{sr}} > 0$ denote the strength-reduction factor, distinct from the Lamé parameter used later in three dimensions. With c_0, ϕ, ψ denoting the unreduced cohesion, friction angle, and dilation angle, respectively, the benchmark uses

$$\begin{aligned} c_1 &= c_0/\lambda_{\text{sr}}, & \phi_1 &= \arctan(\tan\phi/\lambda_{\text{sr}}), & \psi_1 &= \arctan(\tan\psi/\lambda_{\text{sr}}), \\ \beta &= \frac{\cos\phi_1 \cos\psi_1}{1 - \sin\phi_1 \sin\psi_1}, & c_\lambda &= \beta c_1, \\ \phi_\lambda &= \arctan(\beta\tan\phi_1), & \psi_\lambda &= \arctan(\beta\tan\psi_1). \end{aligned} \quad (18)$$

The reported endpoint runs in this paper use $\lambda_{\text{sr}} = 1$ and $\psi = 0°$, hence $\psi_\lambda = 0°$; the reported states are algorithmic endpoint surrogates rather than path-consistent history updates.

The branch potential is then evaluated from the trial state. For a trial engineering strain $\varepsilon = (\varepsilon_{xx}, \varepsilon_{yy}, \gamma_{xy})^\top$ and old plastic strain $\varepsilon^{p,\text{old}}$, set

$$\varepsilon^e = \varepsilon - \varepsilon^{p,\text{old}}, \qquad \sigma^{\text{tr}} = C_{2\text{D}}\varepsilon^e, \qquad S_{2\text{D}} = C_{2\text{D}}^{-1}.$$

Writing $\sigma^{\text{tr}} = (\sigma_{xx}, \sigma_{yy}, \tau_{xy})^\top$, the principal stresses use the smoothing parameter $\epsilon_{\text{reg}} = 1 \times 10^{-12}$:

$$\sigma_{1,2}^{\text{tr}} = m \pm \sqrt{d^2 + \tau_{xy}^2 + \epsilon_{\text{reg}}^2}, \qquad m = \frac{\sigma_{xx} + \sigma_{yy}}{2}, \qquad d = \frac{\sigma_{xx} - \sigma_{yy}}{2}.$$

With $a_\lambda = 1 + \sin\phi_\lambda$, $b_\lambda = 1 - \sin\phi_\lambda$, and $\kappa_\lambda = 2c_\lambda \cos\phi_\lambda$, the trial state is elastic when

$$f^{\text{tr}} = a_\lambda \sigma_1^{\text{tr}} - b_\lambda \sigma_2^{\text{tr}} - \kappa_\lambda \le 0.$$

Thus the elastic branch is the branch for which the trial function in the ordered variables satisfies $f^{\text{tr}} \le 0$.

In that case $\sigma^\star = \sigma^{\text{tr}}$. Otherwise let

$$z = \widehat{S}^{-1}(a_\lambda, -b_\lambda)^\top, \qquad \eta = \frac{f^{\text{tr}}}{a_\lambda z_1 - b_\lambda z_2}, \qquad \widehat{\sigma}^\ell = (\sigma_1^{\text{tr}}, \sigma_2^{\text{tr}})^\top - \eta z,$$

where $\widehat{S} = \begin{bmatrix} S_{11} & S_{12} \\ S_{12} & S_{22} \end{bmatrix}$ is the normal block of $S_{2\text{D}}$ in the ordered principal-stress coordinates. If $\widehat{\sigma}_1^\ell \ge \widehat{\sigma}_2^\ell$, set $\widehat{\sigma}^\star = \widehat{\sigma}^\ell$; otherwise set $\widehat{\sigma}_1^\star = \widehat{\sigma}_2^\star = \kappa_\lambda/(a_\lambda - b_\lambda)$.

Let $\vartheta = \frac{1}{2}\operatorname{atan2}(2\tau_{xy}, \sigma_{xx} - \sigma_{yy})$ be the trial principal direction. We reconstruct the Cartesian stress as

$$\sigma^\star = \begin{bmatrix} \bar{\sigma}^\star + r^\star \cos(2\vartheta) \\ \bar{\sigma}^\star - r^\star \cos(2\vartheta) \\ r^\star \sin(2\vartheta) \end{bmatrix}, \qquad \bar{\sigma}^\star = \frac{\widehat{\sigma}_1^\star + \widehat{\sigma}_2^\star}{2}, \quad r^\star = \frac{\widehat{\sigma}_1^\star - \widehat{\sigma}_2^\star}{2}.$$

The branch potential is

$$\Phi_{\text{MC,2D}} = \sigma^\star \cdot \varepsilon^e - \frac{1}{2}\sigma^\star \cdot S_{2\text{D}}\sigma^\star.$$

The reported two-dimensional rows use this quantity as a solver-diagnostic endpoint surrogate. The small square-root regularization smooths the endpoint surrogate; it is not a separate material parameter.

Using these local definitions, the benchmark uses the following scalarized discrete surrogate functional for differentiation and comparison. Let $c_{\lambda,eq}$ and $\phi_{\lambda,eq}$ denote the quadrature-point Davis-B-reduced cohesion and friction angle:

$$\Pi_h^{2\text{D}}(u_h; \lambda_{\text{sr}}) = \sum_{e=1}^{n_{\text{el}}} \sum_{q=1}^{n_q} w_{eq}\, \Phi_{\text{MC,2D}}\big(B_{eq}u_e, \varepsilon_{eq}^{p,\text{old}}, E, \nu, c_{\lambda,eq}, \phi_{\lambda,eq}\big) - f_{\text{ext}}^\top u_h, \tag{19}$$

Here the subscript eq means element e and quadrature point q, w_{eq} is the corresponding quadrature weight including the element Jacobian, and B_{eq} maps the element displacement vector to the two-dimensional engineering-strain vector used by the branch potential. The load vector f_{ext} is assembled from the right-hand side of (17); for the endpoint solves this is the gravity load with $b = (0, -\gamma)^\top$ and zero traction on the free boundary. The dilation angle enters this endpoint model only through the Davis-B reduction.

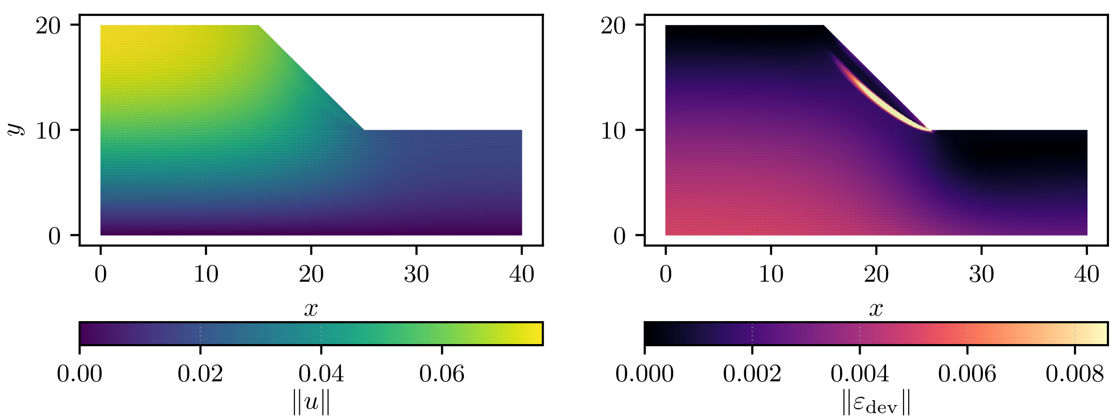


Figure 12: Representative Plasticity2D state: displacement magnitude and deviatoric strain derived from the scalar surrogate (19).

Figure 12 localizes the endpoint response in the expected slope mechanism region. The visualization supports the benchmark definition as an endpoint surrogate; convergence and fixed-work solver evidence are reported separately below.

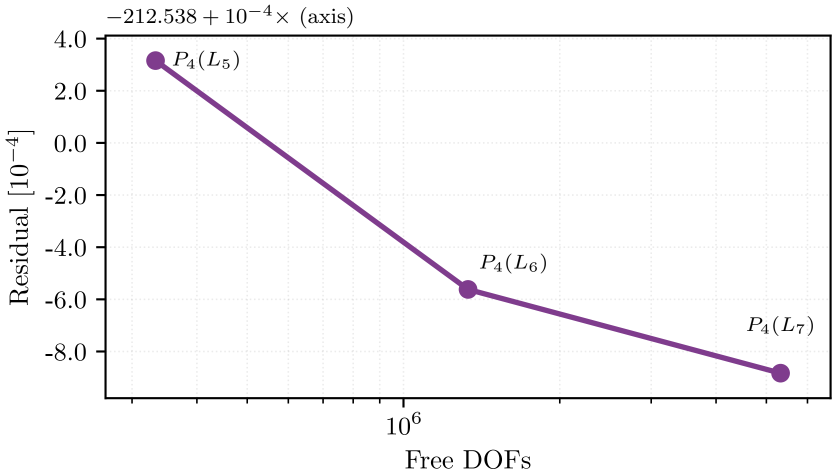


Figure 13: Resolution trend for the Plasticity2D family. The plotted terminal energies are evaluations of (19) on the completed $P_4(L_5)$ endpoint case and on the larger 20-iteration capped $P_4(L_6)$ and $P_4(L_7)$ diagnostics, which are not used as endpoint-convergence evidence.

Table 10: Representative Plasticity2D endpoint and fixed-work diagnostic values from (19).

Case	Free DOFs	Energy	Wall time [s]	Note
$P_4(L_5)$	332,960	−212.538	147	completed endpoint
$P_4(L_6)$	1,331,520	−212.539	147	20-iteration capped diagnostic at 8 ranks
$P_4(L_7)$	5,325,440	−212.539	253	20-iteration capped diagnostic at 16 ranks

This family is where multigrid hierarchy design first becomes dominant in the paper. Only the completed $P_4(L_5)$ endpoint case is treated as an endpoint solve; the larger $P_4(L_6)$ and $P_4(L_7)$ cases are 20-iteration capped diagnostics for solver behavior. Nonlinear and multigrid solver evidence for Plasticity2D is reported later in Sections A and 7.6.

5.5 Plasticity3D

The literature model is again quasi-static Mohr–Coulomb equilibrium with constitutive/history updates, now on a three-dimensional heterogeneous slope geometry [1–5, 26–28]. The unknown is a displacement field $u = (u_x, u_y, u_z) : \Omega \to \mathbb{R}^3$, and gravity acts in the vertical y-direction. Let $e_y = (0, 1, 0)^\top$. Let $\Gamma_{D,i}$ denote the component-wise Dirichlet boundary for displacement component i, let Γ_N be the remaining traction-free boundary, and let V_0 be the homogeneous vector-valued test space after applying these component-wise constraints. With material-region weight $\gamma(x)$, the continuum equilibrium equations are

$$-\nabla\cdot\sigma^{n+1} = -\gamma(x)e_y \quad \text{in } \Omega, \qquad u_i^{n+1} = \bar{u}_i^{n+1} \quad \text{on } \Gamma_{D,i}, \qquad \sigma^{n+1}n = 0 \quad \text{on } \Gamma_N, \tag{20}$$

where σ^{n+1} and the internal variables are obtained from the elastoplastic update driven by $\varepsilon(u^{n+1})$ and the previous state.

The heterogeneous benchmark uses four material regions. Their $(c_0, \phi, \psi, E, \nu, \gamma)$ values, in the order cover layer, general foundation, weak foundation, and slope mass, are $(15, 30^\circ, 0^\circ, 10{,}000, 0.33, 19)$, $(15, 38^\circ, 0^\circ, 50{,}000, 0.30, 22)$, $(10, 35^\circ, 0^\circ, 50{,}000, 0.30, 21)$, and $(18, 32^\circ, 0^\circ, 20{,}000, 0.33, 20)$. At quadrature point q, the body force is $-\gamma_q e_y$, where γ_q is selected by the material region. The elastic constants used below are

$$\mu = \frac{E}{2(1+\nu)}, \qquad K = \frac{E}{3(1-2\nu)}, \qquad \lambda_{\mathrm{L}} = K - \frac{2\mu}{3}.$$

With these regionwise quantities, the weak form used by the endpoint solve is

$$\int_\Omega \sigma^{n+1} : \varepsilon(v)\,\mathrm{d}x = -\int_\Omega \gamma(x)e_y\cdot v\,\mathrm{d}x \qquad \forall v \in V_0. \tag{21}$$

The studied discrete model uses a heterogeneous tetrahedral mesh with four material physical groups, refined into the displayed mesh hierarchy and elevated locally to same-mesh $P_1/P_2/P_4$ spaces. Let $\Gamma_1, \dots, \Gamma_5$ denote the boundary label sets of the reference mesh. In coordinates $X = (X_1, X_2, X_3)$, the mesh bounding box is $[-175, 30] \times [0, 60] \times [0, 86.6025]$ and X_2 is vertical. The labels $\Gamma_1, \Gamma_2, \Gamma_3, \Gamma_4, \Gamma_5$ correspond, respectively, to the right face $X_1 = 30$, the left face $X_1 = -175$, the front face $X_3 = 0$, the back face $X_3 = 86.6025$, and the bottom face $X_2 = 0$; the remaining exterior labels contain the sloped free surface and the top face. We impose

$$\Gamma_{D,x} = \Gamma_1 \cup \Gamma_2, \qquad \Gamma_{D,y} = \Gamma_5, \qquad \Gamma_{D,z} = \Gamma_3 \cup \Gamma_4.$$

The bottom-clamped model additionally fixes all displacement components on $\Gamma_{\text{bot}} = \{X \in \partial\Omega : |X_2| \le 10^{-12}\}$. We write $P_k(L_\ell)$ for polynomial degree k on mesh level ℓ and use $L_1, \ldots, L_4$ for the four displayed mesh levels.

Here and in the Davis-B reduction below, $\lambda_{\text{sr}} > 0$ denotes the strength-reduction factor. The figures below use the bottom-clamped $P_4(L_2)$ representative endpoint at $\lambda_{\text{sr}} = 1.55$ and the finished nine-case study at the same load factor. The $\lambda_{\text{sr}} = 1.0$ strong-scaling study is reported separately in Section 7.7 as auxiliary timing evidence.

We use the engineering-strain ordering

$$\varepsilon = \begin{bmatrix} \varepsilon_{xx} & \varepsilon_{yy} & \varepsilon_{zz} & \gamma_{xy} & \gamma_{yz} & \gamma_{xz} \end{bmatrix}^\top, \tag{22}$$

which is first mapped to tensor shear before principal values are formed. Following the same later strength-reduction descriptions used for the two-dimensional case, the benchmark keeps the Davis-B reduction explicitly at quadrature points:

$$\begin{aligned} c_{1,q} &= c_{0,q}/\lambda_{\text{sr}}, & \phi_{1,q} &= \arctan(\tan\phi_q/\lambda_{\text{sr}})\,, & \psi_{1,q} &= \arctan(\tan\psi_q/\lambda_{\text{sr}})\,, \\ \beta_q &= \frac{\cos\phi_{1,q}\cos\psi_{1,q}}{1-\sin\phi_{1,q}\sin\psi_{1,q}}, & c_{\lambda,q} &= \beta_q c_{1,q}, & \phi_{\lambda,q} &= \arctan(\beta_q \tan\phi_{1,q})\,, \end{aligned} \tag{23}$$

with

$$\bar{c}_{\lambda,q} = 2c_{\lambda,q}\cos\phi_{\lambda,q}, \qquad s_{\lambda,q} := \sin(\phi_{\lambda,q}). \tag{24}$$

These are the scalar constitutive parameters used by the 3D assembly path [1, 3, 26]. In this subsection, ϕ denotes friction angle and ψ denotes dilation angle. All 3D material tuples used in the reported runs have $\psi = 0°$; the dilation angle therefore enters only through the Davis-B reduction above and not as a separate argument of $\Phi_{\text{MC,3D}}$. No separate reduced dilation-angle argument is carried in the 3D branch expressions below.

The local constitutive surrogate follows the piecewise branch structure of the matched endpoint reference formulation. At one quadrature point we suppress the index q and write $s_\lambda := s_{\lambda,q}$, $\bar{c}_\lambda := \bar{c}_{\lambda,q}$, $\mu := \mu_q$, $K := K_q$, and $\lambda_{\text{L}} := \lambda_{\text{L},q}$. Let $\varepsilon_1 \ge \varepsilon_2 \ge \varepsilon_3$ be the ordered principal strains obtained after converting engineering shear components to tensor shear. The branches represent the elastic case and the admissible plastic-return cases of the endpoint surrogate; away from branch interfaces the tests select one smooth branch for differentiation. The branch tests below use the convention that the elastic case satisfies $f_{\text{tr}} \le 0$. Let

$$I_1 = \varepsilon_{xx} + \varepsilon_{yy} + \varepsilon_{zz}, \qquad Q(\varepsilon) = \varepsilon_{xx}^2 + \varepsilon_{yy}^2 + \varepsilon_{zz}^2 + \frac{1}{2}\left(\gamma_{xy}^2 + \gamma_{yz}^2 + \gamma_{xz}^2\right). \tag{25}$$

The elastic branch is

$$\Phi_{\text{el}} = \frac{1}{2}\lambda_{\text{L}} I_1^2 + \mu Q(\varepsilon), \tag{26}$$

and the branch-selection variables are

$$\begin{aligned} f_{\text{tr}} &= 2\mu((1+s_\lambda)\varepsilon_1 - (1-s_\lambda)\varepsilon_3) + 2\lambda_{\text{L}} s_\lambda I_1 - \bar{c}_\lambda, \\ r_{\text{sl}} &= \frac{\varepsilon_1 - \varepsilon_2}{1+s_\lambda}, & r_{\text{sr}} &= \frac{\varepsilon_2 - \varepsilon_3}{1-s_\lambda}, \\ r_{\text{la}} &= \frac{\varepsilon_1 + \varepsilon_2 - 2\varepsilon_3}{3-s_\lambda}, & r_{\text{ra}} &= \frac{2\varepsilon_1 - \varepsilon_2 - \varepsilon_3}{3+s_\lambda}. \end{aligned} \tag{27}$$

The symbols r_{sl}, r_{sr}, r_{la}, and r_{ra} denote return-test ratios; they are kept distinct from the material weight $\gamma(x)$ and the engineering shear components in (22).

The return denominators are

$$\begin{aligned} D_{\text{s}} &= 4\lambda_{\text{L}} s_\lambda^2 + 4\mu(1+s_\lambda^2), & D_{\text{l}} &= 4\lambda_{\text{L}} s_\lambda^2 + \mu(1+s_\lambda)^2 + 2\mu(1-s_\lambda)^2, \\ D_{\text{r}} &= 4\lambda_{\text{L}} s_\lambda^2 + 2\mu(1+s_\lambda)^2 + \mu(1-s_\lambda)^2, & D_{\text{a}} &= 4K s_\lambda^2. \end{aligned} \tag{28}$$

The corresponding branch multipliers are

$$\begin{aligned} \Lambda_{\text{s}} &= \frac{f_{\text{tr}}}{D_{\text{s}}}, \\ \Lambda_{\text{l}} &= \frac{\mu((1+s_\lambda)(\varepsilon_1+\varepsilon_2) - 2(1-s_\lambda)\varepsilon_3) + 2\lambda_{\text{L}} s_\lambda I_1 - \bar{c}_\lambda}{D_{\text{l}}}, \\ \Lambda_{\text{r}} &= \frac{\mu(2(1+s_\lambda)\varepsilon_1 - (1-s_\lambda)(\varepsilon_2+\varepsilon_3)) + 2\lambda_{\text{L}} s_\lambda I_1 - \bar{c}_\lambda}{D_{\text{r}}}, \\ \Lambda_{\text{a}} &= \frac{2K s_\lambda I_1 - \bar{c}_\lambda}{D_{\text{a}}}. \end{aligned} \tag{29}$$

The branch energies are

$$
\begin{aligned}
\Phi_{\rm s} &= \Phi_{\rm el} - \frac{1}{2} D_{\rm s}\Lambda_{\rm s}^2,\\
\Phi_{\rm l} &= \frac{1}{2}\lambda_{\rm L} I_1^2 + \mu\Big(\varepsilon_3^2 + \frac{1}{2}(\varepsilon_1+\varepsilon_2)^2\Big) - \frac{1}{2} D_{\rm l}\Lambda_{\rm l}^2,\\
\Phi_{\rm r} &= \frac{1}{2}\lambda_{\rm L} I_1^2 + \mu\Big(\varepsilon_1^2 + \frac{1}{2}(\varepsilon_2+\varepsilon_3)^2\Big) - \frac{1}{2} D_{\rm r}\Lambda_{\rm r}^2,\\
\Phi_{\rm a} &= \frac{1}{2} K I_1^2 - \frac{1}{2} D_{\rm a}\Lambda_{\rm a}^2.
\end{aligned}
\tag{30}
$$

With these definitions, the scalar potential $\Phi_{\rm MC,3D}(\varepsilon;\bar{c}_\lambda, s_\lambda, \mu, K, \lambda_{\rm L})$ is the following piecewise function. Branch selection is elastic-first: the plastic branches are evaluated only when $f_{\rm tr} > 0$, and ties on branch interfaces are assigned by the first applicable row:

$$
\Phi_{\rm MC,3D}(\varepsilon;\bar{c}_\lambda, s_\lambda, \mu, K, \lambda_{\rm L}) = \begin{cases} \Phi_{\rm el}, & f_{\rm tr} \le 0,\\ \Phi_{\rm s}, & f_{\rm tr} > 0 \text{ and } \Lambda_{\rm s} \le \min(r_{\rm sl}, r_{\rm sr}),\\ \Phi_{\rm l}, & f_{\rm tr} > 0 \text{ and } r_{\rm sl} < r_{\rm sr} \text{ and } r_{\rm sl} \le \Lambda_{\rm l} \le r_{\rm la},\\ \Phi_{\rm r}, & f_{\rm tr} > 0 \text{ and } r_{\rm sr} < r_{\rm sl} \text{ and } r_{\rm sr} \le \Lambda_{\rm r} \le r_{\rm ra},\\ \Phi_{\rm a}, & \text{otherwise}, \end{cases} . \tag{31}
$$

The constitutive-AD path differentiates the smooth branches of (31) directly with JAX rather than substituting a hand-derived return-map tangent; the branch interfaces remain nonsmooth and are treated as part of the algorithmic constitutive surrogate rather than as a globally smooth continuum potential.

For the endpoint and continuation studies, these local definitions are assembled into the following scalar surrogate. Let $\bar{c}_{\lambda,eq}$ and $s_{\lambda,eq}$ denote the quadrature-point Davis-B-reduced quantities:

$$
\Pi_h^{\rm 3D}(u_h;\lambda_{\rm sr}) = \sum_{e=1}^{n_{\rm el}}\sum_{q=1}^{n_q} w_{eq}\,\Phi_{\rm MC,3D}(\varepsilon_{eq}(u_e), \bar{c}_{\lambda,eq}, s_{\lambda,eq}, \mu_{eq}, K_{eq}, \lambda_{{\rm L},eq}) - f_{\rm ext}^\top u_h, \tag{32}
$$

where $\lambda_{\rm sr}$ is again the strength-reduction factor and $\lambda_{\rm L}$ is the Lamé parameter of the elastic predictor. The subscript eq means that each material parameter is evaluated at element e and quadrature point q. The load vector $f_{\rm ext}$ is assembled from the body force $-\gamma_q e_y$ in (21), with no Neumann traction contribution in the reported cases.

For the strain visualizations in this paper we use the deviatoric norm

$$
\|\varepsilon_{\rm dev}\| = \sqrt{\varepsilon^\top \left(\mathrm{diag}(1,1,1,\tfrac{1}{2},\tfrac{1}{2},\tfrac{1}{2}) - \frac{1}{3} m m^\top\right)\varepsilon}, \qquad m = (1,1,1,0,0,0)^\top, \tag{33}
$$

which matches the engineering-shear convention of (22).

Equations (32) and (31) define an algorithmic endpoint surrogate rather than a path-consistent incremental-history functional. The paper therefore keeps Plasticity3D claims at the endpoint-surrogate level and treats direct-branch endpoint observables separately from the fixed-load reference comparison in Section 6.3. The related return-mapping, optimization, convex-optimization, and continuation literature [2–5] provides reference-model context for that boundary.

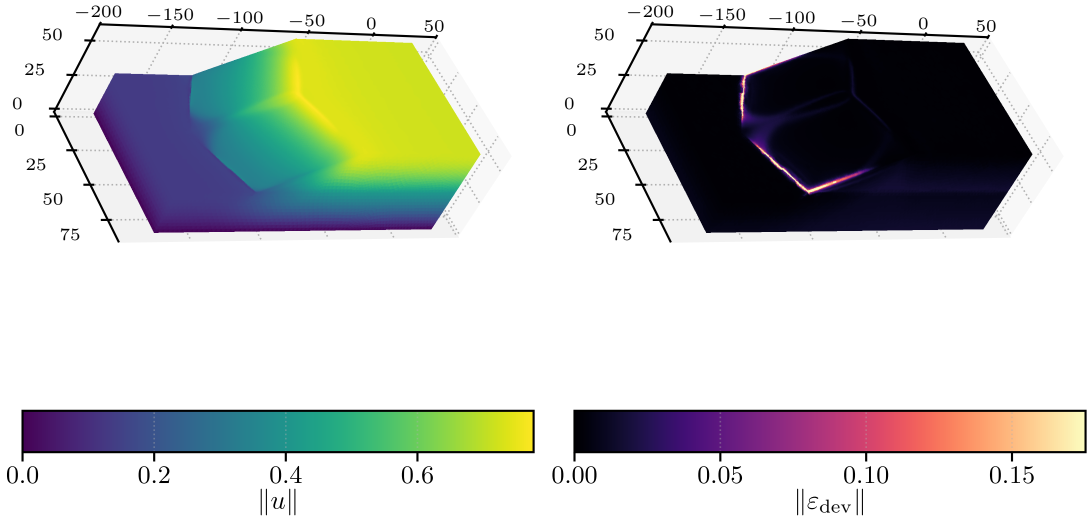


Figure 14: Bottom-clamped Plasticity3D representative endpoint at $\lambda_{\rm sr} = 1.55$: displacement magnitude and surface deviatoric strain for the $P_4(L_2)$ run, both derived from the scalar surrogate in (32).

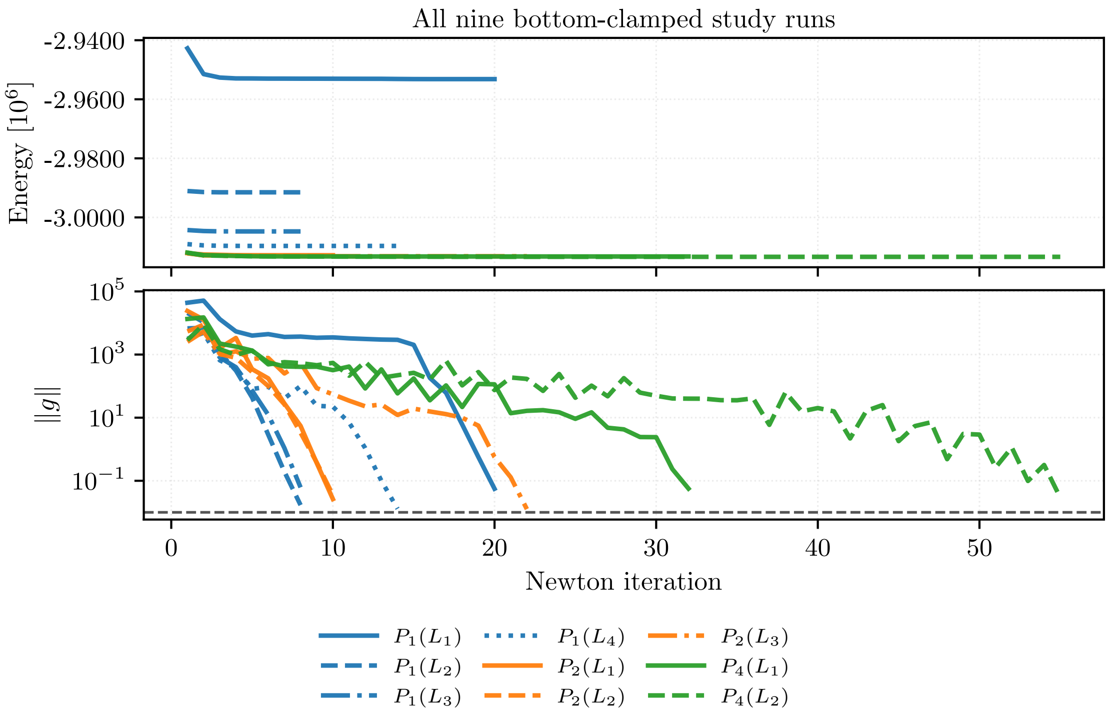


Figure 15: Newton-history convergence for all nine completed bottom-clamped Plasticity3D study runs at $\lambda_{\rm sr} = 1.55$. The upper panel shows the value of the algorithmic surrogate from (32); the lower panel shows the gradient norm used as the nonlinear stopping metric, with color indicating degree and line style indicating mesh level.

Table 11: All nine completed bottom-clamped Plasticity3D study cases at $\lambda_{\mathrm{sr}} = 1.55$, reported as endpoint values of (32).

Element	Free DOFs	Energy	$\|g\|_{\mathrm{final}}$	Wall time [s]
$P_1(L_1)$	10,526	−2,953,167	4.56×10^{-3}	1.79
$P_1(L_2)$	79,024	−2,991,497	1.64×10^{-3}	8.72
$P_1(L_3)$	610,964	−3,004,742	4.07×10^{-3}	17.2
$P_1(L_4)$	4,801,816	−3,009,658	1.16×10^{-3}	96.7
$P_2(L_1)$	79,024	−3,012,813	2.47×10^{-3}	18.3
$P_2(L_2)$	610,964	−3,013,032	2.96×10^{-3}	98.3
$P_2(L_3)$	4,801,816	−3,013,149	1.14×10^{-3}	1330
$P_4(L_1)$	610,964	−3,013,227	5.53×10^{-3}	208
$P_4(L_2)$	4,801,816	−3,013,348	4.42×10^{-3}	2298

Figures 14 and 15 and table 11 define the endpoint-surrogate benchmark object used later in the results section. The nine completed endpoint solves converge in the nonlinear stopping metric while the terminal values of (32) decrease along both mesh and degree refinements.

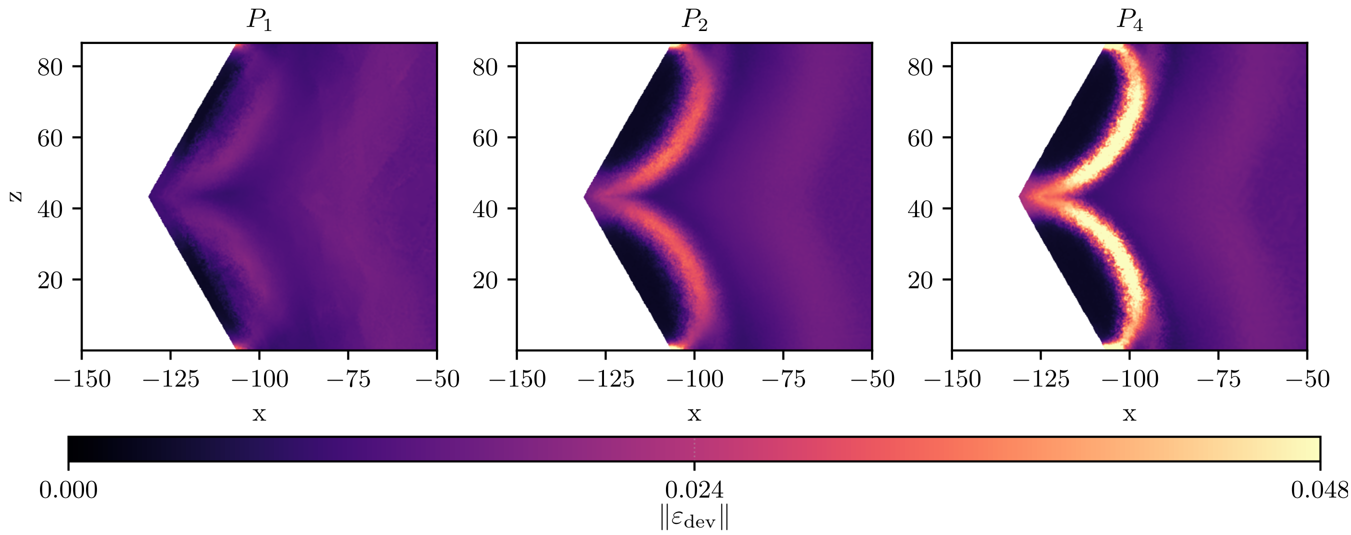


Figure 16: Shared-scale upper y-slice comparison of $\|\varepsilon_{\text{dev}}\|$ from (33) at the highest mesh level of each degree line: $P_1(L_4)$, $P_2(L_3)$, and $P_4(L_2)$. The three runs use the bottom-clamped solver configuration at $\lambda_{\text{sr}} = 1.55$ and are compared on one common color scale.

Figure 16 adds a shared-scale slice comparison. The larger values of $\|\varepsilon_{\text{dev}}\|$ remain localized near the same upper slope region across P_1, P_2, and P_4, so the later degree and timing comparisons are interpreted as endpoint-solver comparisons rather than changes of the physical problem.

Plasticity3D is the benchmark where detailed constitutive definitions are most important: the equations above define the endpoint-surrogate quantities reported in the results section, namely (32) and (33). Timing, scaling, and degree-versus-resolution evidence for this family are reported in Section 7.7.

5.6 Topology Optimization

The topology benchmark tests the toolset on a coupled design–mechanics loop rather than on a single equilibrium minimization. The literature setting is the standard SIMP-style compliance topology-optimization problem on a cantilever domain, together with filtering and continuation ideas from the classical topology literature [19–22]. In the setting studied here, each outer iteration solves a linear elasticity problem for the current density and then minimizes a frozen reduced design objective. The pure JAX topology realization is only a compact serial design demonstration; the agreement evidence for this family is the JAX+PETSc rank-consistency check in Section 7.8.

The reported fine-grid run uses $\Omega = (0, 2) \times (0, 1)$ with a 768×384 structured triangular mesh, target volume fraction 0.4, $\theta_{\min} = 1 \times 10^{-6}$, Young's modulus $E = 1$, and Poisson ratio 0.3. The left edge is clamped, and a downward unit traction acts on the load patch $\{2\} \times [0.4, 0.6]$. With $h_x = 2/768$ and $h_y = 1/384$, fixed-solid design pads are imposed on $x \le 32h_x$ and on $x \ge 2 - 32h_x$, $y \in [0.4 - 32h_y, 0.6 + 32h_y]$.

The first ingredient is the design map. Let Z_h be the scalar continuous P_1 space on the mesh. The design variable is a latent nodal field $z_h \in Z_h$, mapped to a physical density through the logistic transform

$$\theta_h(z_h) = \theta_{\min} + (1 - \theta_{\min})\,\mathrm{sigmoid}(z_h), \qquad \mathrm{sigmoid}(z) = \frac{1}{1 + e^{-z}}. \tag{34}$$

On each triangular element e, we use the element averages $\theta_e := |e|^{-1} \int_e \theta_h \,\mathrm{d}x$ and $z_e := |e|^{-1} \int_e z_h \,\mathrm{d}x$; for the linear elements used here these are vertex averages. On the fixed-solid pads the physical density is held at $\theta_e = 1$ during design updates. Outside the pads, the latent variable is unconstrained and (34) enforces $\theta_h \in [\theta_{\min}, 1]$.

The second ingredient is the mechanics solve for a fixed density field. Let V_h be the continuous vector P_1 displacement space and let $V_{h,0} \subset V_h$ impose the homogeneous left-edge clamp. Let C denote the plane-stress elasticity tensor and let p be the active SIMP exponent. For fixed density, define

$$a_{\theta_h}(w_h, v_h) := \sum_{e=1}^{n_{\text{el}}} |e|\, \theta_e^p \left(\varepsilon(w_h)|_e\right)^\top C \left(\varepsilon(v_h)|_e\right), \tag{35}$$

where $\varepsilon(v) = (\partial_x v_x, \partial_y v_y, \partial_y v_x + \partial_x v_y)^\top$, and

$$C = \frac{E}{1 - \nu^2} \begin{bmatrix} 1 & \nu & 0 \\ \nu & 1 & 0 \\ 0 & 0 & (1-\nu)/2 \end{bmatrix}.$$

The load functional is

$$\ell(v) = \int_{\{2\}\times[0.4,0.6]} (0,-1)\cdot v \,\mathrm{d}s.$$

The mechanics step finds $u_h(\theta_h)$ in the affine displacement space such that $a_{\theta_h}(u_h, v_h) = \ell(v_h)$ for all $v_h \in V_{h,0}$. The scalar compliance observable reported below is

$$\mathcal{C}_h(\theta_h) := \ell(u_h(\theta_h)) = a_{\theta_h}(u_h(\theta_h), u_h(\theta_h)).$$

It is an observable of the mechanics solve, whereas (36) is the frozen design subproblem solved at one outer iteration.

The third ingredient is the reduced design update. At outer iteration m, after solving for $u_h^m = u_h(\theta_h^m)$, we freeze the elementwise quantity

$$e_{e,m}^{\text{frozen}} = (\theta_e^m)^{2p}\, \varepsilon(u_h^m)^\top C \varepsilon(u_h^m).$$

This is a reciprocal local compliance approximation. The mechanics stiffness in (35) is proportional to θ_e^p, so $e_{e,m}^{\text{frozen}}\theta_e^{-p}$ matches the current local compliance scale at $\theta_e = \theta_e^m$ while penalizing lower trial densities. Let $(\nabla\theta_h)|_e$ denote the elementwise gradient of the interpolated density field used in the phase-field term, and define the double-well density

$$W_{\text{dw}}(\theta) := \theta^2(1-\theta)^2.$$

The effective volume multiplier λ_V^m is defined below, and α_{reg}, ℓ_{pf}, and μ_{move} are, respectively, the phase-field weight, phase-field length scale, and move regularization parameter. The reduced objective for the next design vector z_h^{m+1} is

$$\mathcal{J}_h^m(z_h) = \sum_{e=1}^{n_{\text{el}}} |e| \left[e_{e,m}^{\text{frozen}}\theta_e^{-p} + \lambda_V^m \theta_e + \alpha_{\text{reg}}\left(\frac{\ell_{\text{pf}}}{2}|(\nabla\theta_h)|_e|^2 + \frac{1}{\ell_{\text{pf}}} W_{\text{dw}}(\theta_e)\right) + \frac{\mu_{\text{move}}}{2}(z_e - z_e^m)^2 \right], \tag{36}$$

where θ_e and z_e are computed from the trial z_h, while z_e^m is the element average from the previous design iterate.

The target volume fraction is imposed through an iteration-dependent effective multiplier. The fixed-solid pads are included in the measured volume, and the normalized volume is

$$V(\theta_h) := \frac{1}{|\Omega|}\sum_{e=1}^{n_{\text{el}}} |e|\,\theta_e.$$

Let $V_\star = 0.4$ be the target value. Before the mth design solve we use

$$r_V^m = V(\theta_h^m) - V_\star, \qquad \lambda_V^m = \lambda_{\text{ref}}^m + \lambda_{\text{corr}}^m + \rho_V r_V^m,$$

where λ_{ref}^m is the $(1-V_\star)$-quantile of $p\, e_{e,m}^{\text{frozen}} / \max(\theta_e^m, \theta_{\min})^{p+1}$ over the design elements, λ_{corr}^m is the carried multiplier correction, and $\rho_V = 10$. After the design update, with $r_V^{m+1} = V(\theta_h^{m+1}) - V_\star$, the carried correction is updated by

$$\lambda_{\text{corr}}^{m+1} = \lambda_{\text{corr}}^m + \beta_V \max(1, \lambda_{\text{ref}}^m) r_V^{m+1}, \qquad \beta_V = 12.$$

Here λ_V^m is the effective volume multiplier in the current design subproblem. The SIMP-style continuation in p and the regularization terms control mesh-scale design features, following the role of filters in Bourdin [22]. The constants $\alpha_{\text{reg}} = 5\times10^{-3}$, $\ell_{\text{pf}} = 8\times10^{-2}$, and $\mu_{\text{move}} = 1\times10^{-2}$ set the phase-field and latent-move terms. These are algorithmic choices of this benchmark, not standard SIMP notation. The outer loop alternates mechanics solves for (35) with reduced design updates for (36).

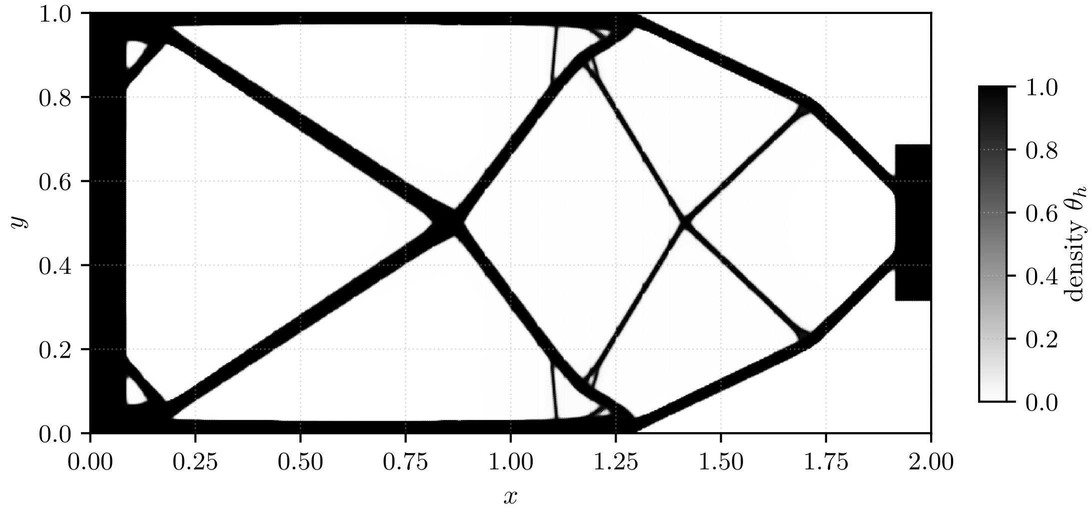


Figure 17: Representative final density field for the parallel topology benchmark.

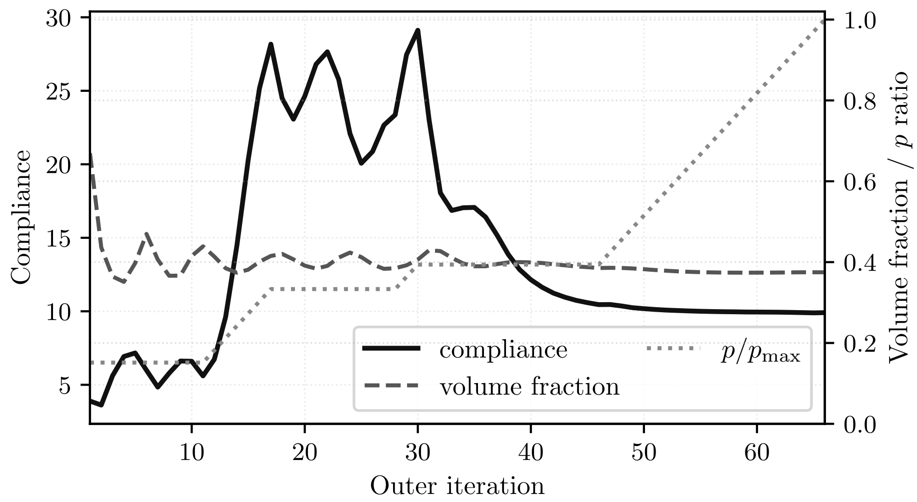


Figure 18: Objective-history view for the reduced topology objective (36), with compliance on the left axis and volume fraction plus the continuation schedule in p on the right axis.

Table 12: Representative topology endpoint observables for the reduced-design run.

Case	Ranks	Outer iters	Compliance	Volume fraction	Wall time [s]
192×96	1	121	4.1557	0.4	10
768×384	32	66	9.9074	0.3749	25.1

Figures 17 and 18 and Table 12 show the reduced-design endpoint used for the topology timing study. The fine-grid run produces a connected load-bearing density field, reaches volume fraction 0.3749, and stops after 66 outer iterations under the continuation logic shown in Figure 18. The target volume is imposed through the adaptive multiplier and stopping rule, so the endpoint volume is reported as an achieved value rather than as an exact equality-constraint satisfaction claim. For topology, the relevant notion of "convergence" is this reduced objective history rather than a single equilibrium energy. The reported continuation starts at SIMP exponent $p = 1$, increases p by 0.2 on each outer iteration up to $p = 10$, and treats the fine-grid run as stalled once $p \geq 4$ and both the reported design-change and state-change stopping diagnostics are at most 10^{-6}. Rank-varied adaptive timing and a controlled rank-consistency check for the same family are reported in Section 7.8.

6 Validation and External Comparisons

This section separates validation and external-comparison evidence from the performance results that follow. The distinction is important because the paper uses different levels of evidence for different claims: Hyperelasticity has a narrow external comparison against JAX-FEM, while Plasticity3D is evaluated through a matched endpoint-comparator formulation for the surrogate studied here.

6.1 Comparison Metrics

For vector and field comparisons with nonzero reference norm, we report relative differences of the form

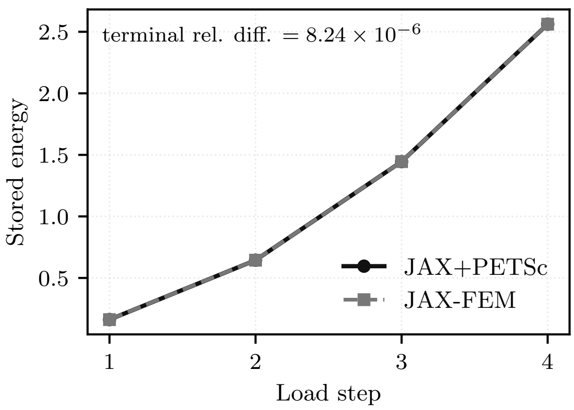
$$\delta_M(a,b) = \frac{\|a-b\|_M}{\|a\|_M}, \tag{37}$$

where a is the reference observable, b is the compared observable, and M denotes the discrete norm used for the comparison. For nodal displacement vectors, M is the Euclidean norm on the compared free degrees of freedom. For sampled profiles and quadrature-derived fields, M denotes the corresponding sampled or quadrature-weighted norm used by the comparison. The reported field and profile observables have nonzero reference norms. Scalar work and energy differences use the same reference-denominator convention; the hyperelastic energy post-comparison uses an absolute denominator floor of 10^{-12}. We write $u_{\max}$ for the maximum displacement magnitude over the compared state.

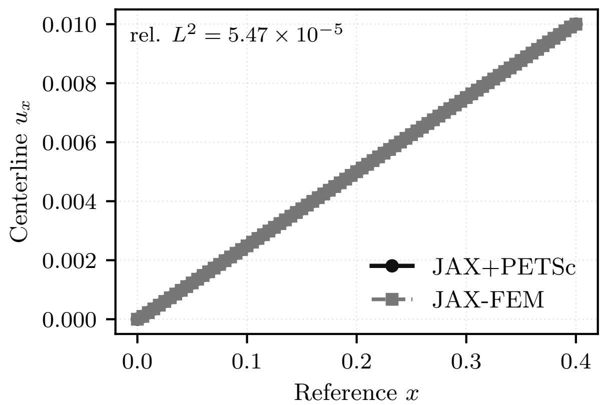
The hyperelastic JAX-FEM comparison uses a 5% predefined threshold for the final energy, Euclidean nodal full-field displacement, centerline displacement, and $u_{\max}$-curve differences. The Plasticity3D fixed-load matched-comparator diagnostic uses three predefined criteria: at most 0.03 relative difference in the highest successful load factor, at most 0.05 relative L^2 difference for the $u_{\max}(\lambda_{\mathrm{sr}})$ curve, and at most 0.10 relative L^2 difference for the endpoint displacement. The deviatoric-strain and boundary-profile differences in Table 14 are diagnostics, not acceptance criteria. These thresholds are predeclared engineering agreement gates for companion and surrogate comparisons. They are not literature validation tolerances and do not replace path-history validation for elastoplastic evolution.

6.2 Hyperelasticity External Comparison

The hyperelasticity energy is defined in Section 5.3. The external comparison here is deliberately narrow and uses a serial companion case rather than the 24-step rotating benchmark path. Both implementations use the same level-1 mesh and the same right-face x-translation schedule $d = (0.0025, 0.005, 0.0075, 0.01)$, then post-compare both terminal states with the energy in (12). This supports terminal-state observable agreement for this hyperelastic companion problem only. The comparison is therefore an energy-postprocessed terminal-state check for one hyperelastic case, and it does not assert constitutive-law identity between the two implementations: the specific JAX-FEM companion case used here uses a logarithmic-J Piola form, whereas (12) uses the $D_1(J-1)^2$ volumetric penalty.



(a) Terminal energy over the matched displacement schedule.



(b) Final centerline u_x profile on the common sample path.

Figure 19: Serial hyperelastic comparison against JAX-FEM on the same mesh and prescribed displacement schedule, with energy post-comparison of the terminal states under (12).

Table 13: Thresholded terminal-state companion comparison against JAX-FEM, with energy post-comparison under (12).

Group	Quantity	Relative difference	Status
Agreement	final energy	8.24×10^{-6}	below 5%
Agreement	full-field displacement relative Euclidean	2.01×10^{-4}	below 5%
Agreement	centerline relative L^2	5.47×10^{-5}	below 5%
Agreement	$u_{\max}$ curve relative L^2	8.85×10^{-16}	below 5%
Condition	common mesh	–	satisfied
Condition	common displacement schedule	–	satisfied
Condition	agreement threshold (5%)	–	satisfied
Condition	energy re-evaluation	–	applied

Figure 19 shows that the terminal-energy history and centerline displacement profile remain close on the matched schedule. The table quantifies that visual agreement under the common post-comparison energy.

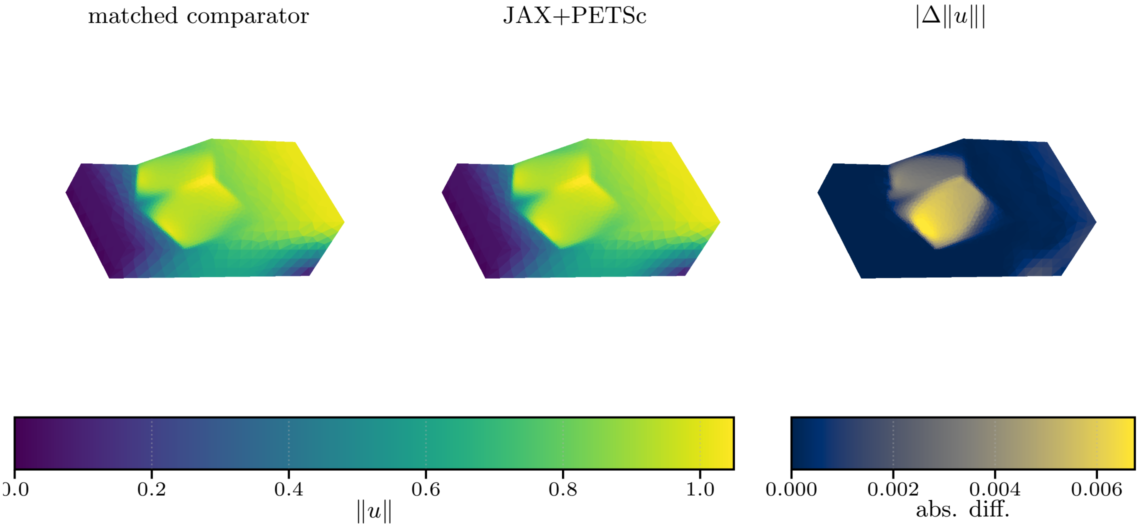
The agreement criteria in Table 13 are satisfied for all checked quantities. The final energy relative difference is 8.244×10^{-6}, the full-field displacement relative Euclidean nodal difference is 2.011×10^{-4}, the centerline relative L^2 difference is 5.473×10^{-5}, and the $u_{\max}$-curve relative L^2 difference is effectively zero. The scientific role of this table is terminal-state agreement under a matched mesh, displacement schedule, and common energy post-comparison for this companion hyperelastic case.

6.3 Plasticity3D Endpoint-Surrogate Comparison

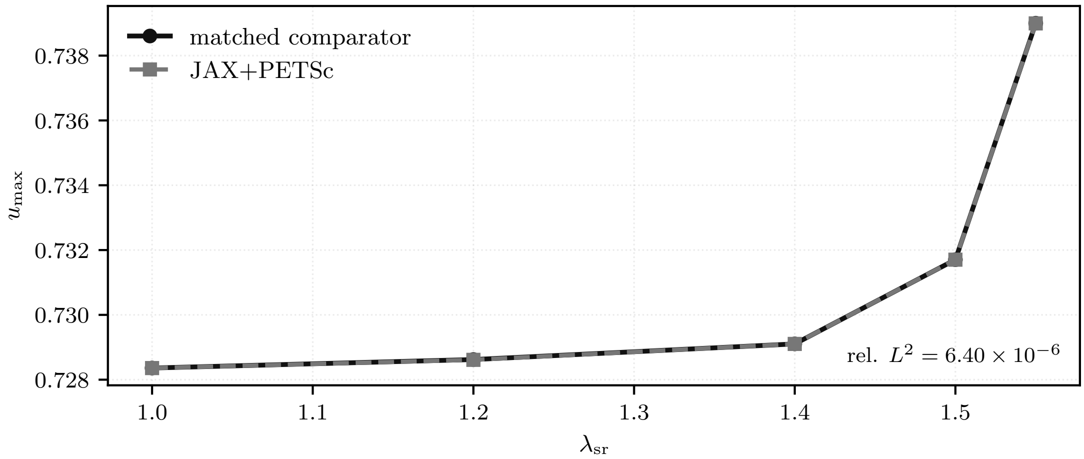
The Plasticity3D benchmark is tied to Mohr–Coulomb slope-stability reference formulations, but the object studied here is the algorithmic endpoint surrogate defined in Section 5.5 by (32) and (31). The Sysala return-mapping, optimization, convex-optimization, and advanced continuation papers provide reference-model and method context for this problem class [2–5]. The efficient Čermák–Sysala–Valdman MATLAB elastoplastic implementation of Čermák et al. [18] provides implementation context for practical two- and three-dimensional elastoplastic finite-element solvers. None of these sources is used here as a verified numerical baseline.

The endpoint-surrogate comparison therefore uses two diagnostics. Here the matched endpoint comparator means the endpoint formulation used for these observable comparisons, not the cited incremental-history studies. The comparator uses the same endpoint functional, mesh, material table, Davis-B strength reduction, load schedule, boundary conditions, and active free degrees of freedom, but evaluates the endpoint observables through an alternate matched route. The endpoint-observable check tests a seven-step direct load branch, $\lambda_{\mathrm{sr}} = 1.0, 1.1, \dots, 1.6$, against the matched endpoint comparator on the $P_2(L_1)$ comparator case. The work observable is the external work $\omega = f_{\mathrm{ext}}^{\top} u$ reported by the two endpoint formulations. Its relative difference is 3.877×10^{-5}, the displacement relative L^2 difference is 3.517×10^{-3}, and the deviatoric-strain relative L^2 difference is 8.720×10^{-3}.

The fixed-load matched-comparator diagnostic asks a narrower fixed-λ_{sr} question against the matched endpoint comparator. After aligning the comparator boundary conditions with the active boundary condition set, the fixed-load grid is capped at the main-figure load level $\lambda_{\mathrm{sr}} = 1.55$. Here $\lambda_{\max}^{\mathrm{succ}}$ denotes the largest value in the tested grid $\{1.0, 1.2, 1.4, 1.5, 1.55\}$ that satisfies the relative-correction stopping criterion. The comparison matches $\lambda_{\max}^{\mathrm{succ}}$ on that grid and satisfies the predefined criteria for the $u_{\max}(\lambda_{\mathrm{sr}})$ curve and endpoint displacement: the $u_{\max}(\lambda_{\mathrm{sr}})$ relative L^2 difference is 6.396×10^{-6}, and the endpoint displacement relative L^2 difference is 8.299×10^{-4}. The endpoint deviatoric-strain relative L^2 difference (4.307×10^{-3}) and boundary-profile relative L^2 difference (5.229×10^{-4}) are reported as diagnostics rather than acceptance criteria. This remains an endpoint check rather than a path-consistent plastic-history reference.



(a) Endpoint-observable comparison with the matched endpoint comparator.



(b) Fixed-λ_{sr} matched-comparator diagnostic.

Figure 20: Plasticity3D endpoint-surrogate comparison. The upper panel compares the endpoint observables after the seven-step direct branch ending at $\lambda_{\mathrm{sr}} = 1.6$; the lower panel shows the fixed-λ_{sr} matched-comparator diagnostic after matching the boundary conditions and active free degrees of freedom.

Table 14: Plasticity3D endpoint-surrogate comparison summary for the direct-branch endpoint-observable comparison ending at $\lambda_{\mathrm{sr}} = 1.6$ and the fixed-λ_{sr} matched-comparator diagnostic.

Check	Comparison	Relative difference	Status
endpoint observable	work	3.88×10^{-5}	comparison
endpoint observable	displacement relative L^2	3.52×10^{-3}	comparison
endpoint observable	deviatoric-strain relative L^2	8.72×10^{-3}	comparison
fixed-load comparison	$\lambda_{\max}^{\mathrm{succ}}$	0	satisfied
fixed-load comparison	$u_{\max}(\lambda_{\mathrm{sr}})$ relative L^2	6.4×10^{-6}	satisfied
fixed-load comparison	endpoint displacement relative L^2	8.3×10^{-4}	satisfied
fixed-load comparison	endpoint deviatoric-strain relative L^2	4.31×10^{-3}	diagnostic
fixed-load comparison	boundary profile relative L^2	5.23×10^{-4}	diagnostic
fixed-load comparison	fixed-load criteria, 3/3 satisfied	–	satisfied

Figure 20 shows the two endpoint-surrogate diagnostics used for the comparison: the direct-branch observable check and the fixed-λ_{sr} curve after boundary-condition matching. The resulting interpretation is limited but useful. Plasticity3D is reported here through direct-branch endpoint-observable differences, while the fixed-load matched-comparator diagnostic supports matched endpoint observables for that surrogate under matched boundary conditions and active free degrees of freedom. Neither check is a path-consistent incremental-plasticity history validation.

7 Performance and Solver Behavior

This section turns from problem definition to computational behavior. The benchmark families in Section 5 defined the governing equations, discrete energies, and representative states, and Section 6 separated the validation and external-comparison evidence. Here we focus on parallel timings, scaling trends, solver behavior, and the comparative cost of alternative derivative or discretization choices. Earlier timing evidence is retained when it clarifies scalability, but it is treated as performance evidence rather than validation evidence. The cross-section protocol table below recalls validation rows only to fix the terminology used when those rows are discussed alongside timing evidence. The status and timing terms used below follow the protocol in Table 2.

Table 15: Numerical evidence blocks and reporting protocol. Validation rows are recalled only for terminology; the performance interpretations below use the timing, stopping, and solver-policy columns. The linear-solve column gives the representative KSP tolerance and maximum-iteration policy; result-specific rows override it when stated.

Evidence block	Role	Solver policy	KSP rtol / max iters	Stop or fixed work	Reported time
Scalar benchmarks	formulation, derivative-route, and globalization evidence	Newton with Armijo or the stated trust-region variant; CG or GMRES with Hypre in PETSc rows	p-Laplace: $10^{-1}/30$; Ginzburg–Landau: $10^{-3}/200$	configured nonlinear tolerances; capped rows are diagnostic	wall or solve/elapsed time in globalization rows
Hyperelasticity	finite-strain mechanics, JAX-FEM comparison, and scaling	trust-region or line-search Newton; GAMG or same-mesh PMG profiles as stated	GAMG rows: $10^{-1}/30$; PMG rows stated with diagnostics	per-load-step stationarity; fixed-step rows are solver diagnostics	wall time for benchmark rows; solver total for first-step scaling
Plasticity2D	endpoint evidence plus larger fixed-work diagnostics	Armijo continuation with a same-mesh PMG hierarchy	typically $10^{-2}/15$ unless a diagnostic states otherwise	$P_4(L_5)$ is the endpoint solve; $P_4(L_6)$–$P_4(L_7)$ are fixed-iteration diagnostics	wall time for the stated endpoint or fixed-work run
Plasticity3D validation	endpoint-surrogate agreement, not path-history validation	fixed-load matched-comparator diagnostics under matched boundary conditions	not the primary timing target	validation criteria use fixed-load observables; strain/profile rows are diagnostic	timing is secondary to agreement metrics
Plasticity3D performance	second-order routes, globalization, and parallel scaling	Armijo, residual-bisection, or trust-region Newton with FGMRES and same-mesh PMG	10^{-1} or 10^{-2} with max 100, as stated by row	relative correction, gradient target, cap, or fixed work as stated	wall, solve, or solver-total time according to the table header
Topology	distributed design-mechanics timing and rank consistency	adaptive reduced-objective continuation with PETSc mechanics and GAMG	mechanics FGMRES/GAMG: $10^{-4}/100$	design/state-change stall criterion or fixed schedule	end-to-end wall time

7.1 Globalization Method Comparison

Table 16 asks whether one nonlinear globalization policy is robust across the tested families. Each row holds the element assembly path and preconditioner family fixed within the problem and varies only the nonlinear policy. Timed-out rows are retained as capped solver behavior rather than convergence evidence.

The rows use representative fixed cases rather than one common physical problem. The p-Laplace row is a separate unit-load stress test on the level-10 hierarchy, the hyperelasticity row uses the first eight load steps of the level-4 rank-local assembly path, and the Plasticity3D row uses the $P_2(L_1)$ endpoint surrogate at $\lambda_{\text{sr}} = 1.55$. The p-Laplace, Ginzburg–Landau, hyperelasticity, and Plasticity3D rows use 32, 16, 32, and 32 MPI ranks, respectively, with wall-time caps of 30, 120, 270, and 180 s.

The scalar and mechanics rows show different preferences. Newton with Armijo line search is fastest on the clean p-Laplace unit-load row, but the same line-search-only policy times out on Ginzburg–Landau. For the hyperelasticity excerpt all three policies complete; the hybrid policy reduces nonlinear iterations and solver time relative to line-search-only Newton, while the pure Steihaug variant has more trust-region rejections.

The Plasticity3D endpoint gives the strongest warning against a universal policy. At the stricter $\|\nabla\mathcal{F}\|_2 < 10^{-4}$ criterion, line-search-only Newton reaches the iteration cap, whereas both trust-region policies converge. The capped row matches the displayed energy but remains above the gradient target, as reported by the gradient-to-target column. The hybrid variant is the faster trust-region configuration and also uses fewer Krylov iterations. These comparisons support a problem-dependent policy choice rather than a universal preference for any single globalization strategy.

We additionally evaluate the Ginzburg–Landau level-10 case under an eight-rank budget. Table 17 reports this scalar-nonconvex comparison separately from the multi-family table above. Its purpose is narrower: it records that the trust-region variants complete with matching energy and Newton–Krylov work, while the line-search-only configuration still times out under the fixed wall-time cap. For that timed-out configuration we report elapsed wall time and the cap, whereas completed configurations report solver time.

7.2 Derivative-Route Comparison

Table 18 asks whether the second-order route changes the terminal state or mainly changes the assembly and linearization cost. The three matched cases compare element AD, colored sparse recovery, and, where available, constitutive AD under the same nonlinear problem definition. The timing-scope column states whether the reported time is a solver total, solve time, or wall time for the specific configuration.

Table 16: Fixed-case comparison of nonlinear globalization policies. LS denotes Armijo line search and TR denotes Steihaug trust region. The gradient-to-target column is reported for the Plasticity3D endpoint gate.

Outcome and terminal value

Benchmark	Method	Ranks	Outcome	Completed/requested steps	Solve/elapsed [s]	Grad/target	Energy
p-Laplace L_{10}	Newton + LS	32	completed	1/1	6.95	–	−0.2517224496
p-Laplace L_{10}	Steihaug TR	32	completed	1/1	11.7	–	−0.2517227312
p-Laplace L_{10}	Hybrid TR+LS	32	completed	1/1	12	–	−0.2517227312
Ginzburg–Landau L_{10}	Newton + LS	16	timeout	0/1	121	–	–
Ginzburg–Landau L_{10}	Steihaug TR	16	completed	1/1	14.3	–	0.3456246783
Ginzburg–Landau L_{10}	Hybrid TR+LS	16	completed	1/1	14.3	–	0.3456246783
Hyperelasticity L_4	Newton + LS	32	completed	8/8	194	–	9.733145
Hyperelasticity L_4	Steihaug TR	32	completed	8/8	178	–	9.73101
Hyperelasticity L_4	Hybrid TR+LS	32	completed	8/8	149	–	9.730978
Plasticity3D $P_2(L_1)$	Newton + LS	32	iteration cap	0/1	70.2	1.18	−3,012,813
Plasticity3D $P_2(L_1)$	Steihaug TR	32	completed	1/1	20.8	0.00967	−3,012,813
Plasticity3D $P_2(L_1)$	Hybrid TR+LS	32	completed	1/1	16.4	0.579	−3,012,813

Nonlinear and Krylov work

Benchmark	Method	Ranks	Newton	Krylov	LS evals	TR rejects
p-Laplace L_{10}	Newton + LS	32	15	32	15	0
p-Laplace L_{10}	Steihaug TR	32	29	32	0	0
p-Laplace L_{10}	Hybrid TR+LS	32	29	32	29	0
Ginzburg–Landau L_{10}	Newton + LS	16	0	0	0	0
Ginzburg–Landau L_{10}	Steihaug TR	16	19	25	0	0
Ginzburg–Landau L_{10}	Hybrid TR+LS	16	19	25	19	0
Hyperelasticity L_4	Newton + LS	32	420	6984	2758	0
Hyperelasticity L_4	Steihaug TR	32	314	7245	0	95
Hyperelasticity L_4	Hybrid TR+LS	32	295	6849	499	3
Plasticity3D $P_2(L_1)$	Newton + LS	32	80	858	224	0
Plasticity3D $P_2(L_1)$	Steihaug TR	32	13	96	0	1
Plasticity3D $P_2(L_1)$	Hybrid TR+LS	32	11	73	12	0

Table 17: Ginzburg–Landau level-10 fixed-budget globalization comparison on eight MPI ranks. The configurations use the same element assembly and Hypre preconditioner; only the nonlinear globalization policy changes. The timed-out configuration reports elapsed wall time and the configured wall-time cap.

Method	Outcome	Newton	Krylov	LS evals	TR rejects	Solve/elapsed [s]	Cap [s]	Energy
Newton + LS	timeout	–	–	–	–	301	300	–
Steihaug TR	completed	19	27	0	0	28.5	–	0.3456246783
Hybrid TR+LS	completed	19	27	19	0	28.5	–	0.3456246783

Across all three benchmark cases, colored sparse recovery reaches the same table-reported energy, up to displayed precision, and identical nonlinear/Krylov iteration counts as element AD. The scalar and hyperelasticity cases show a wall-time premium for colored sparse recovery. On the Plasticity3D endpoint, constitutive AD is fastest, while element AD and colored sparse recovery have matching iteration counts and comparable solve times. The colored sparse-recovery route requires 240–420 colors across the 32 ranks; its comparable solve time is consistent with batched rank-local HVPs having nearly the same measured Hessian-evaluation cost as element AD on this $P_2(L_1)$ case.

Table 19 adds the degree sensitivity behind the Plasticity3D derivative-route claim. This single-linearization comparison uses one Newton linearization on bottom-clamped heterogeneous meshes at $\lambda_{\text{sr}} = 1.55$, 32 MPI ranks, and the degree-appropriate PMG hierarchy. The table therefore measures Hessian-evaluation cost at fixed linearization, not nonlinear convergence quality. The same-DOF comparison is the most informative same-DOF contrast: $P_1(L_2)$ and $P_2(L_1)$ both have 79,024 free unknowns, but the P_2 element stencil is denser. Its colored sparse-recovery route therefore needs 240–420 colors instead of 66–90, raises the Hessian evaluation from 0.332 to 1.412 s, and raises the rank-maximum RSS from about 0.65 to 1.10 GiB. The P_4 colored sparse-recovery case is outside this single-linearization table; the P_2 same-DOF comparison already isolates the stencil-density effect.

Table 18: Matched-case derivative-route comparison. Colored SFD denotes rank-local colored sparse recovery; the reported configurations use AD-HVP probes, while finite-difference gradient probes are the classical recovery variant. The SFD-color range is shown where per-rank color counts are available.

Outcome and timing

Benchmark	Route	Outcome	Time [s]	Timing scope	Energy
p-Laplace L_9	Element AD	completed	1.14	solver total	−7.960003
p-Laplace L_9	Colored SFD	completed	1.45	solver total	−7.960003
Hyperelasticity L_4 step 1	Element AD	completed	11.7	solve	0.1519923275
Hyperelasticity L_4 step 1	Colored SFD	completed	34.6	solve	0.1519923275
Plasticity3D $P_2(L_1)$ $\lambda_{sr} = 1.55$	Element AD	completed	16.3	solve	−3,012,813
Plasticity3D $P_2(L_1)$ $\lambda_{sr} = 1.55$	Colored SFD	completed	16.4	solve	−3,012,813
Plasticity3D $P_2(L_1)$ $\lambda_{sr} = 1.55$	Constitutive AD	completed	4.81	solve	−3,012,813

Work and Hessian construction

Benchmark	Route	Ranks	Newton	Krylov	Hessian [s]	SFD colors
p-Laplace L_9	Element AD	32	6	11	–	–
p-Laplace L_9	Colored SFD	32	6	11	–	–
Hyperelasticity L_4 step 1	Element AD	32	22	492	–	–
Hyperelasticity L_4 step 1	Colored SFD	32	22	492	–	–
Plasticity3D $P_2(L_1)$ $\lambda_{sr} = 1.55$	Element AD	32	11	73	6.57	–
Plasticity3D $P_2(L_1)$ $\lambda_{sr} = 1.55$	Colored SFD	32	11	73	6.59	240–420
Plasticity3D $P_2(L_1)$ $\lambda_{sr} = 1.55$	Constitutive AD	32	11	73	1.6	–

Table 19: Plasticity3D one-Newton derivative-route cost versus discretization. Elem DOFs is the local vector element size used by the Hessian evaluation; overlap DOFs is the maximum local overlap vector size recorded during assembly.

Discretization size

Element	Route	Work	Free DOFs	Elem. DOFs	Overlap DOFs	RSS [GiB]
$P_1(L_1)$	Element AD	one Newton step	10,526	12	675	0.5
$P_1(L_1)$	Colored SFD	one Newton step	10,526	12	675	0.5
$P_1(L_1)$	Constitutive AD	one Newton step	10,526	12	675	0.5
$P_1(L_2)$	Element AD	one Newton step	79,024	12	3624	0.6
$P_1(L_2)$	Colored SFD	one Newton step	79,024	12	3624	0.6
$P_1(L_2)$	Constitutive AD	one Newton step	79,024	12	3624	0.5
$P_2(L_1)$	Element AD	one Newton step	79,024	30	3933	0.9
$P_2(L_1)$	Colored SFD	one Newton step	79,024	30	3933	1.1
$P_2(L_1)$	Constitutive AD	one Newton step	79,024	30	3933	0.6
$P_4(L_1)$	Element AD	one Newton step	610,964	105	27177	1.2
$P_4(L_1)$	Constitutive AD	one Newton step	610,964	105	27177	1.1

Linearization cost

Element	Route	Free DOFs	Krylov	Solve [s]	Hessian [s]	Colors
$P_1(L_1)$	Element AD	10,526	11	0.0806	0.257	–
$P_1(L_1)$	Colored SFD	10,526	11	0.0704	0.256	57–90
$P_1(L_1)$	Constitutive AD	10,526	11	0.0706	0.217	–
$P_1(L_2)$	Element AD	79,024	3	0.232	0.322	–
$P_1(L_2)$	Colored SFD	79,024	3	0.273	0.332	66–90
$P_1(L_2)$	Constitutive AD	79,024	3	0.164	0.257	–
$P_2(L_1)$	Element AD	79,024	5	0.83	1.31	–
$P_2(L_1)$	Colored SFD	79,024	5	0.827	1.41	240–420
$P_2(L_1)$	Constitutive AD	79,024	5	0.304	0.535	–
$P_4(L_1)$	Element AD	610,964	13	8.48	7.04	–
$P_4(L_1)$	Constitutive AD	610,964	13	3.97	3.62	–

7.3 p-Laplace

The p-Laplace problem definition is given in Section 5.1, with discrete energy (7). In this clean scalar setting, the numerical story is easiest to interpret: exact element Hessians remain effective in the reported distributed runs, while colored SFD reaches nearly identical discrete energies at higher wall time.

In the finest plotted case, the JAX+PETSc element-AD path reports 1.14 s on 32 ranks, while colored SFD reports 1.45 s with the same Newton and Krylov iteration counts. For the matched 32-rank case summarized in Table 18, the reported energies differ by about 8×10^{-9}, so this block mainly measures the overhead of sparse recovery on a smooth scalar problem. Across the plotted ranks, element AD drops from 10.13 s on one rank to 1.14 s on 32 ranks;

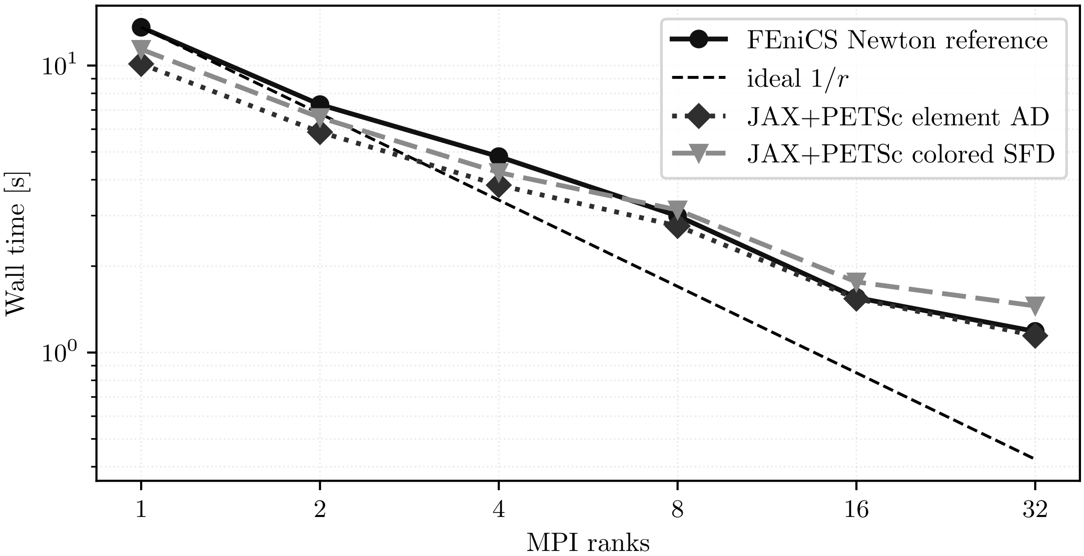


Figure 21: p-Laplace strong scaling on the finest tested grid. All curves correspond to the same discrete problem from (7).

colored SFD follows the same trend with persistent recovery overhead. The FEniCS curve in the same figure provides reference-implementation context; the comparison above isolates the two JAX+PETSc derivative routes.

7.4 Ginzburg–Landau

The Ginzburg–Landau family from Section 5.2, with discrete problem (11), shows the same high-level derivative story under a more indefinite linearization. The important computational message is not a dramatic separation between exact and approximate second-order routes, but that the distributed solver converges in the reported nonconvex scalar-energy runs with the same sparse infrastructure.

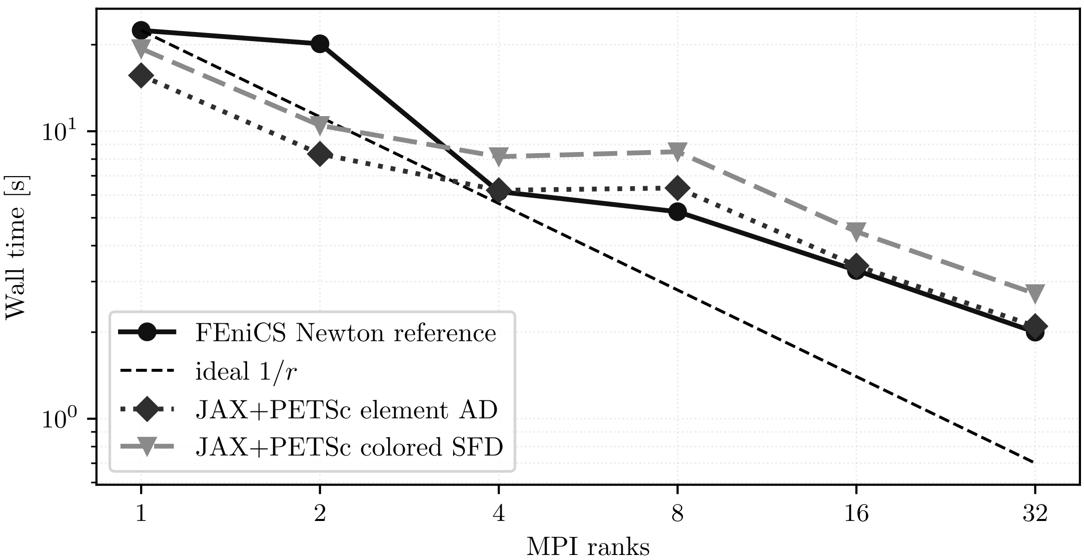


Figure 22: Ginzburg–Landau strong scaling for the tested fine-grid case governed by (11).

At 32 ranks, the element-AD and colored sparse-recovery JAX+PETSc runs report 2.10 s and 2.72 s, respectively. The scaling figure shows the same final-energy trend for the two routes, so the main visible difference is the colored sparse-recovery wall-time premium on this nonconvex scalar energy. Across the plotted ranks, element AD falls from 15.64 s on one rank to 2.10 s on 32 ranks, with a nonideal plateau around 4–8 ranks before the 32-rank point. The FEniCS curve is included as reference-implementation context, while the paragraph above isolates the JAX+PETSc derivative-route cost.

7.5 Hyperelasticity

The hyperelasticity problem definition and boundary data are given in Section 5.3, with discrete stationarity problem (15). In the full load-path evidence, the main solver-policy questions are globalization and preconditioning; the derivative-route comparison is limited to the first-step replicated-element case. The distributed implementation completes the full load path and preserves the terminal energy trend from Section 5.3. The external JAX-FEM comparison is reported separately in Section 6.2; the figure below reports only distributed scaling for the load path in (15).

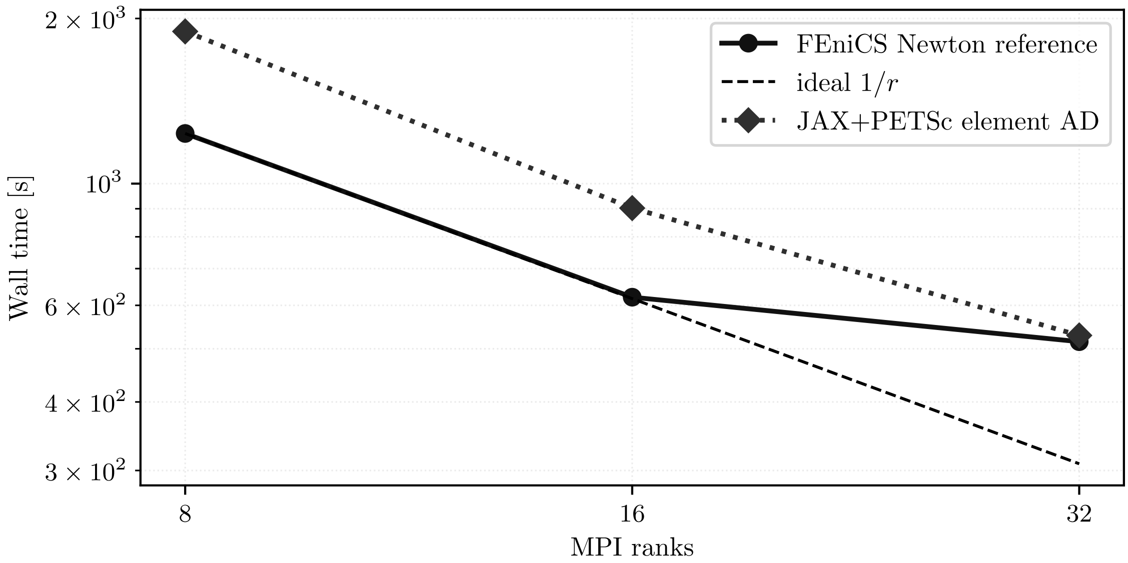


Figure 23: Hyperelasticity strong scaling for the 24-step load path defined by (15).

The 24-step hyperelasticity scaling curve reduces JAX+PETSc element-AD wall time from 1894 s at 8 ranks to 529 s at 32 ranks. The speedup is substantial but not ideal, which motivates the more targeted fixed-work distribution and PMG diagnostics below.

The larger multi-node experiment asks a fixed-work solver-policy question: how the rank-local hyperelasticity assembly behaves when the level-5 first load step uses the same nonlinear work but different rank counts and PMG coarse-solver layouts. The CPU runs use one MPI rank per core on nodes with two AMD EPYC 7H12 64-core processors and 256 GB RAM [39], with contiguous CPU binding and NUMA-local memory binding. The mesh is generated from the uniform beam hierarchy, and the element assembly uses rank-local elements with PETSc COO preallocation and point-to-point overlap exchange.

The nonlinear solve is the same trust-region Newton path used for the benchmark in section 5.3. The linear subproblems use STCG with a same-hierarchy PMG preconditioner, Chebyshev/Jacobi smoothing, and coarsest level L_2. The coarse solve is PETSc redundant LU with MUMPS: the partial one-node configurations use one redundant group over the active ranks, while the multi-node configurations use one 128-rank redundant group per node. Figure 24 and table 22 report measured solver total time rather than scheduler elapsed time. All configurations reach the same nonlinear and linear iteration counts shown in Table 22: 21 Newton iterations and 43 Krylov iterations.

Table 20 reports the memory evidence that supports using the rank-local assembly variant in the larger CPU sweeps. The fixed-work agreement rows compare the replicated and rank-local level-4 first-step solve at four MPI ranks; the one-linearization memory rows use one Newton linearization on the level-5 mesh at 8, 16, and 32 ranks. The table reports energy, operating-system peak RSS, and the measured footprint of assembly arrays, making clear whether a configuration stores global mesh and adjacency data or only local elements plus overlap.

The level-4 fixed-work rows show matching Newton and Krylov counts and the same rounded energy value for replicated and rank-local assembly. On the larger level-5 one-linearization rows, rank-local assembly reduces summed RSS from 215.6 GiB to 104.9 GiB at 32 ranks, with solve times 3.37 s and 3.36 s. The level-5 energy values are fixed-work terminal states after one linearization, so these rows are memory/cost diagnostics rather than endpoint-agreement checks.

Table 21 isolates the first-step level-5 PMG choice on the same 32-rank level-5 problem with fixed nonlinear work. All configurations perform the same eight Newton linearizations, so the table compares the cost and terminal state reached under different preconditioner and coarse-solver policies. The energy column is the state after this fixed nonlinear work, so Table 21 is a solver-policy comparison rather than an equal-accuracy timing comparison or endpoint-accuracy ranking.

Table 20: Rank-local hyperelasticity memory and overlap evidence. RSS max/sum gives the maximum and summed per-rank resident set size. Tracked sum is the reported assembly-array footprint, and overlap/owned is the summed rank-local overlap DOF count divided by global owned free DOFs for the rank-local overlap assembly.

Solve work

Purpose	Assembly layout	Outcome	Level	Ranks	Newton	Krylov	Solve [s]	Energy
fixed-work agreement	replicated	completed	L_4	4	26	571	51.2	0.151971
fixed-work agreement	rank-local	completed	L_4	4	26	571	50.8	0.151971
one-linearization memory	replicated	single linearization	L_5	8	1	1	10.7	53.95804
one-linearization memory	rank-local	single linearization	L_5	8	1	1	10.5	53.95926
one-linearization memory	replicated	single linearization	L_5	16	1	1	5.57	32.927343
one-linearization memory	rank-local	single linearization	L_5	16	1	1	5.5	32.927867
one-linearization memory	replicated	single linearization	L_5	32	1	1	3.37	23.942407
one-linearization memory	rank-local	single linearization	L_5	32	1	1	3.36	23.803027

Memory and overlap

Purpose	Assembly layout	Level	Ranks	RSS max/sum [GiB]	Tracked sum [GiB]	Overlap/owned
fixed-work agreement	replicated	L_4	4	3.6/14.2	5.49	1.01
fixed-work agreement	rank-local	L_4	4	3.3/13.2	6.01	1.01
one-linearization memory	replicated	L_5	8	14.9/118.7	44.34	1.01
one-linearization memory	rank-local	L_5	8	12.3/98.3	47.87	1.01
one-linearization memory	replicated	L_5	16	9.4/150.9	45.45	1.03
one-linearization memory	rank-local	L_5	16	6.3/100.3	47.89	1.03
one-linearization memory	replicated	L_5	32	6.7/215.6	47.65	1.05
one-linearization memory	rank-local	L_5	32	3.3/104.9	47.92	1.05

Table 21: Rank-local hyperelasticity level-5 PMG and coarse-solver sensitivity at 32 MPI ranks with fixed nonlinear work. The PMG configurations use the same rank-local element path as the scaling runs.

Outcome and solver work

Precond.	Nonlinear work	Newton	Krylov	Solver [s]	Energy
GAMG	8 Newton linearizations	8	42	22.3	1.608441
PMG L_2 + Hypre	8 Newton linearizations	8	92	23.3	0.5252349427
PMG L_2 + MUMPS	8 Newton linearizations	8	17	15.3	0.3848912708
PMG L_3 + MUMPS	8 Newton linearizations	8	17	16.3	0.3864166034

Time breakdown and coarse solve

Precond.	Coarse	Assembly [s]	PC [s]	Linear [s]
GAMG	–	9.11	6.41	3.91
PMG L_2 + Hypre	Hypre	10.3	2.93	8.14
PMG L_2 + MUMPS	MUMPS, one redundant group	9.13	2.55	1.91
PMG L_3 + MUMPS	MUMPS, one redundant group	9.14	2.93	2.51

The fixed-work table reports that the MUMPS-coarse PMG variants reduce total Krylov iterations from 92 with Hypre coarse solves to 17, and reduce solver time from 23.3 s to about 16 s. This is preconditioner-policy evidence for a fixed nonlinear workload, not a statement about final load-path accuracy. The energy differences in the table are fixed-work terminal-state differences after the same eight Newton linearizations.

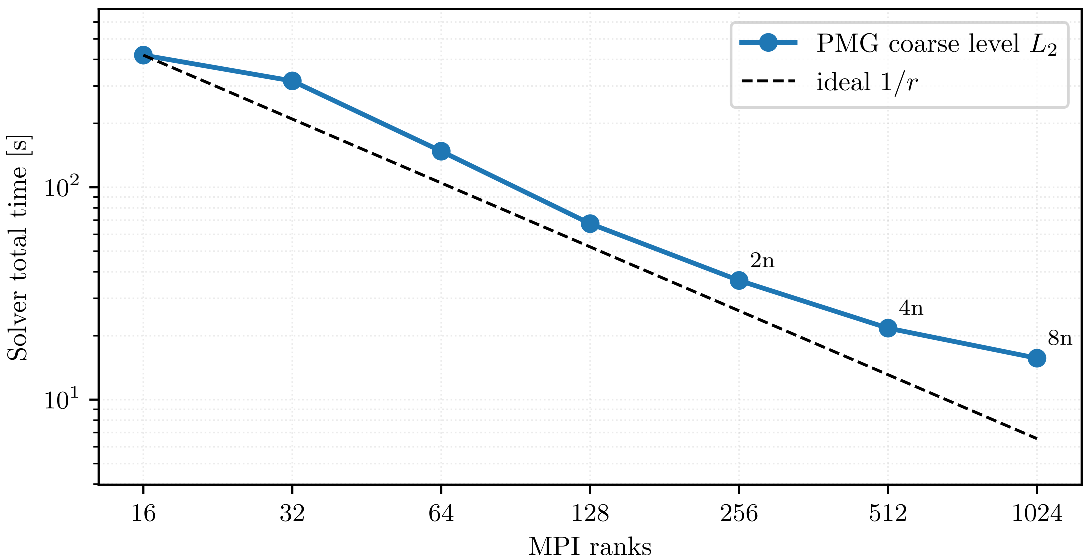


Figure 24: Single- and multi-node CPU level-5 hyperelasticity first-step scaling at fixed nonlinear work with the PMG coarse level fixed at L_2. Annotations show CPU-node counts.

Table 22: Single- and multi-node CPU level-5 hyperelasticity first-step timings used in figure 24.

Ranks	Nodes	Coarse groups	Ranks/group	Solver total [s]	First step [s]	Newton iters	Krylov iters
16	1	1	16	419	367	21	43
32	1	1	32	317	280	21	43
64	1	1	64	148	132	21	43
128	1	1	128	67.4	58.4	21	43
256	2	2	128	36.4	30.7	21	43
512	4	4	128	21.7	17.7	21	43
1024	8	8	128	15.7	12.3	21	43

Figure 24 and table 22 show a fixed-work timing drop from 419 s on 16 ranks to 15.7 s on 1024 ranks, with the same 21 Newton iterations and 43 Krylov iterations in all listed configurations. This evidence isolates the multi-node sparse-solver and PMG-coarse-level behavior for one load step, rather than the accuracy of the full load path.

7.6 Plasticity2D

The Plasticity2D model is defined in Section 5.4, with Davis-B reduction (18) and discrete surrogate (19). For this family, the most informative computational evidence in the present paper is not a single strong-scaling curve. The benchmark section separates the completed endpoint case from the larger fixed-cost studies in Figure 13 and table 10, and the reference-continuation comparison is collected later in Section A.

The $P_4(L_5)$ case is the endpoint evidence for this family. The completed endpoint has energy -212.538 and wall time 147 s. The larger $P_4(L_6)$ and $P_4(L_7)$ cases are fixed-work diagnostics for nonlinear-tail and PMG behavior; both report energy -212.539 after 20-iteration caps, with wall times 147 s and 253 s.

The 8-rank and 32-rank reference-continuation rows in Table 31 support solver-policy timing under a common continuation logic. They are not a separate validation result or a strong-scaling claim. In this diagnostic, increasing from 8 to 32 ranks reduces wall time from 489 s to 198 s, while the continuation Krylov totals differ, 733 versus 829.

7.7 Plasticity3D

The governing problem is defined in Section 5.5, with Davis-B reduction (23), scalar constitutive potential (31), discrete surrogate (32), and deviatoric-strain norm (33). The validation limits for this endpoint surrogate are reported in Section 6.3. With that interpretation fixed, this section asks two performance questions: which second-order route is fastest on the fixed high-order derivative-route case under matched terminal observables, and how the endpoint surrogate behaves across degree, resolution, and rank count.

Derivative-route comparison. The first question is whether the second-order route changes the terminal observables or only the cost of derivative construction. On the fixed $P_4(L_1)$ case at $\lambda_{\mathrm{sr}} = 1.5$ and 8 MPI ranks, Figure 25 and table 23 show identical Newton counts, total Krylov counts, and terminal observables across the three routes. In the table, ω denotes the external-work observable and $u_{\max}$ denotes the maximum displacement magnitude.

The differences are cost differences. Constitutive AD has the lowest reported wall time on this fixed case, with wall time 222 s and solve time 191 s, compared with 288 s/255 s for element AD and 372 s/253 s for colored SFD. This residual-bisection Newton comparison uses the Hypre-backed rank-local linear-solver profile; it is therefore separate from the MUMPS-backed PMG configurations used by the later $\lambda_{\mathrm{sr}} = 1.55$ convergence studies.

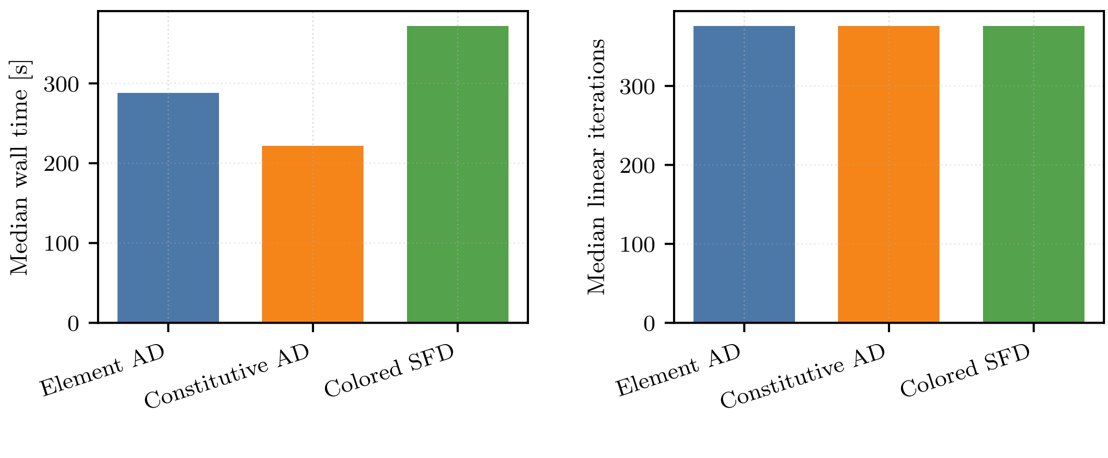


Figure 25: Fixed Plasticity3D derivative-route comparison on $P_4(L_1)$, $\lambda_{\mathrm{sr}} = 1.5$, and 8 MPI ranks.

Table 23: Plasticity3D derivative-route comparison on the fixed $P_4(L_1)$, $\lambda_{\rm sr} = 1.5$, 8-rank case. These configurations use the Hypre-backed rank-local profile and are separate from the MUMPS-backed PMG configurations in the $\lambda_{\rm sr} = 1.55$ studies.

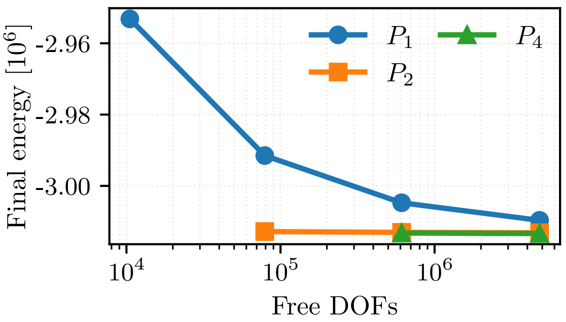
Timing and work

Route	Wall time [s]	Solve time [s]	Newton iters	Krylov iters
Element AD	288	255	8	376
Constitutive AD	222	191	8	376
Colored SFD	372	253	8	376

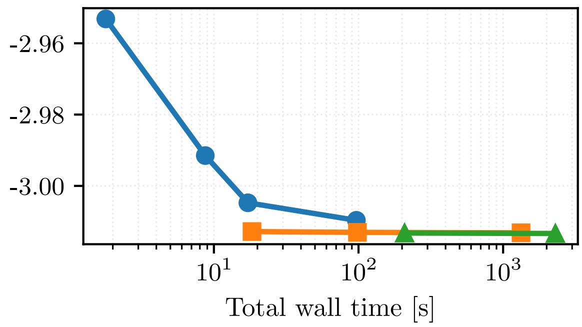
Terminal observables

Route	Energy	ω	$u_{\max}$
Element AD	−3,010,861	6,027,722	0.731848
Constitutive AD	−3,010,861	6,027,722	0.731848
Colored SFD	−3,010,861	6,027,722	0.731848

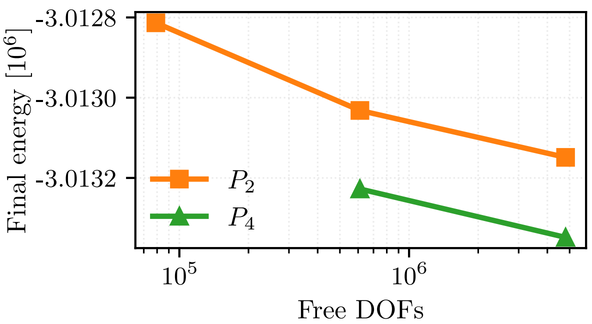
Discretization and scaling. The remaining figures in this subsection answer a different question: how the endpoint surrogate behaves across degree/resolution choices, and how the selected solver configurations scale on the separate $P_4(L_2)$ timing studies. The zoomed panels in Figure 26 are included because the full figure is dominated by coarse P_1 variation. Under the bottom-clamped constraint, the matched set at 610,964 free DOFs is $P_1(L_3)$, $P_2(L_2)$, and $P_4(L_1)$. The matched set at 4,801,816 free DOFs is $P_1(L_4)$, $P_2(L_3)$, and $P_4(L_2)$. The scientific message from Figure 26 and table 11 is therefore that the solver reaches similar terminal energies along both p- and h-refined paths, while higher order on coarser meshes buys lower observed values of (32) at matched DOFs only by paying a clear wall-time premium in this nine-case endpoint study. At 610,964 DOFs, for example, $P_4(L_1)$ reports energy −3,013,227 and wall time 208 s, compared with −3,013,032 and 98.3 s for $P_2(L_2)$. At 4,801,816 DOFs, $P_4(L_2)$ reports −3,013,348 and 2298 s, compared with −3,013,149 and 1330 s for $P_2(L_3)$.



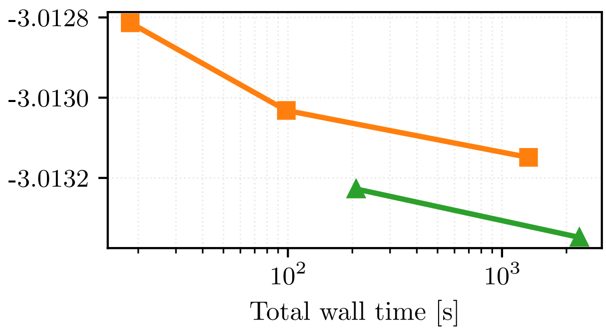
(a) All degrees: final energy versus free DOFs.



(b) All degrees: final energy versus total wall time.



(c) P_2/P_4 zoom: final energy versus free DOFs.



(d) P_2/P_4 zoom: final energy versus total wall time.

Figure 26: Bottom-clamped Plasticity3D degree-versus-resolution study at $\lambda_{\rm sr} = 1.55$. The top panels show the full $P_1/P_2/P_4$ comparison, while the bottom panels zoom to P_2/P_4 so the near-overlap hidden by the coarse P_1 points becomes visible. The left column plots final values of (32) against free DOFs; the right column plots the same converged energies against end-to-end wall time. All timings use the same 32-rank configuration.

Table 24: Separate Plasticity3D strong-scaling study for constitutive-AD rank-local assembly with same-mesh PMG on $P_4(L_2)$ at $\lambda_{sr} = 1.0$. These timings provide solver evidence for a separate load factor from the bottom-clamped $\lambda_{sr} = 1.55$ benchmark figures.

Ranks	Wall time [s]	Solve time [s]	Speedup	Efficiency	Newton iters	Krylov iters
1	5424	5176	1	1	17	114
2	2914	2777	1.861	0.931	17	114
4	1717	1634	3.16	0.79	17	114
8	1164	1107	4.658	0.582	17	114
16	608	575	8.916	0.557	17	114
32	300	284	18.053	0.564	17	114

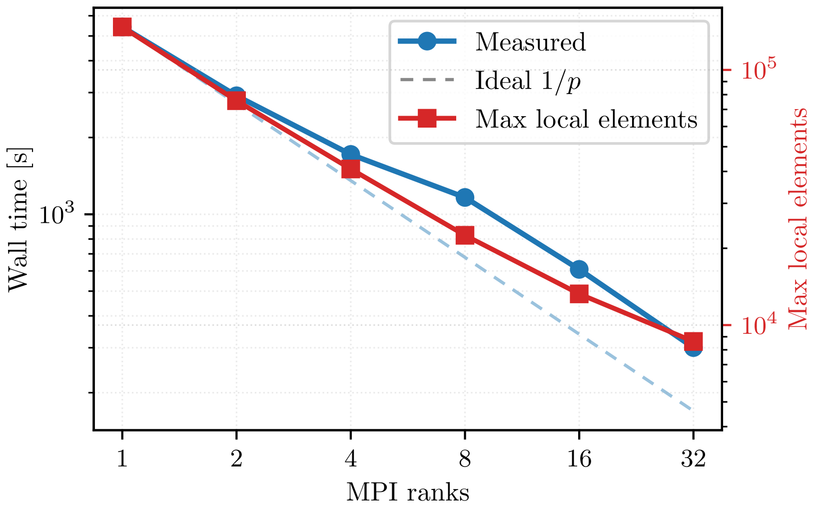


Figure 27: Separate strong-scaling study for the selected Plasticity3D solver configuration: constitutive-AD rank-local assembly with same-mesh PMG on $P_4(L_2)$ at $\lambda_{sr} = 1.0$. This is a separate timing study from the bottom-clamped $\lambda_{sr} = 1.55$ results shown in figures 14, 15 and 26.

Figure 27 and table 24 give a high-DOF Plasticity3D distributed timing result for the Hypre-coarse PMG variant. This scaling evidence uses the selected $P_4(L_2)$ mesh family but a different load factor, $\lambda_{sr} = 1.0$, so it complements rather than extends the bottom-clamped $\lambda_{sr} = 1.55$ benchmark figures above. Within that timing study, the reported nonlinear and linear iteration counts remain matched on 1/2/4/8/16/32 ranks, while the end-to-end time drops from 5424 s on one rank to 300 s on 32 ranks. Additional reference-formula assembly and continuation evidence is reported later in Section A.

The converged bottom-clamped $P_4(L_2)$, $\lambda_{sr} = 1.55$ scaling run keeps the nonlinear work fixed: all plotted configurations converge in 29 Newton iterations and 1240 total Krylov iterations to the same energy and stopping-gradient norm to the precision shown in Table 25. The run uses the same case and load level as the benchmark table, with constitutive-AD rank-local assembly and the MUMPS-backed same-mesh $P_4 \to P_2 \to P_1$ PMG hierarchy. The nonlinear method is Armijo Newton, the PETSc relative KSP residual tolerance is 1×10^{-2}, and the nonlinear stop requires both relative correction below 0.002 and a stopping-gradient norm below one percent of the first Newton gradient. The stopping contract is reported through both $\|g\|_{\text{stop}}$ and $\|g\|_{\text{stop}}/\|g\|_{\text{target}}$; all listed ratios are below one. Figure 28 and table 25 therefore compare single-node and 16-ranks-per-node CPU timings without changing the nonlinear workload.

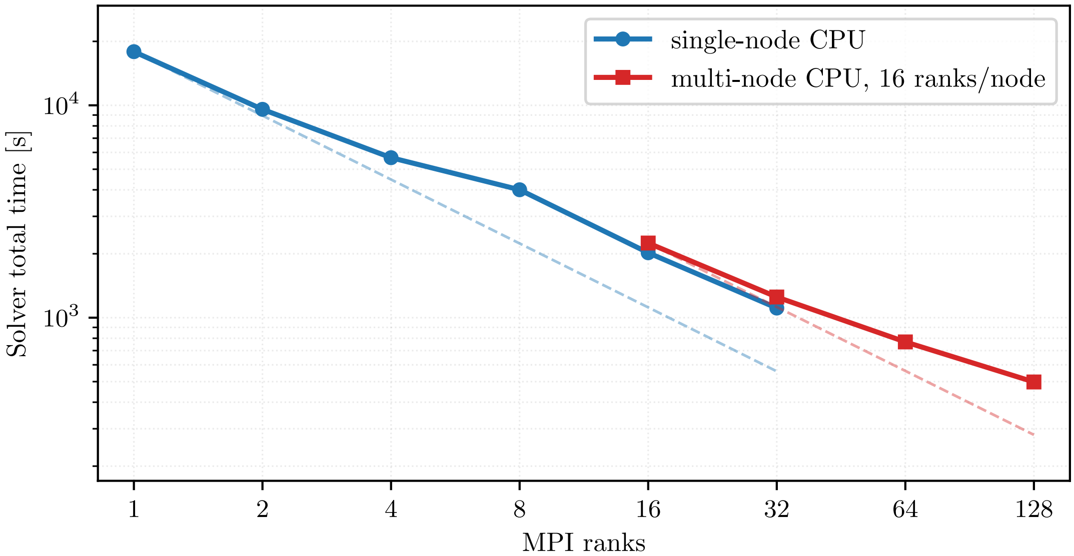


Figure 28: Single-node and multi-node CPU converged scaling for the bottom-clamped Plasticity3D $P_4(L_2)$ case at $\lambda_{sr} = 1.55$. Dashed references are ideal $1/r$ lines anchored separately to the first point of each CPU setting.

Table 25: Converged timings for the Plasticity3D $P_4(L_2)$, $\lambda_{sr} = 1.55$ run in figure 28. Single-node configurations use one CPU node; multi-node configurations use 16 ranks per CPU node. The nonlinear stop requires relative correction below 0.002 and final gradient below one percent of the first Newton gradient. The stopping-gradient norm in this table is the relative-gradient stopping metric for this scaling run, not the absolute final-gradient column in the degree benchmark table.

CPU setting	Ranks	Nodes	Solver total [s]	Solve [s]	Newton iters	Krylov iters	Energy	$\|g\|_{\text{stop}}$	$\|g\|_{\text{stop}}$/target
single-node CPU	1	1	17,890	17,615	29	1240	−3,013,348	10.594	0.525
single-node CPU	2	1	9574	9412	29	1240	−3,013,348	10.594	0.525
single-node CPU	4	1	5667	5564	29	1240	−3,013,348	10.594	0.525
single-node CPU	8	1	4003	3947	29	1240	−3,013,348	10.594	0.525
single-node CPU	16	1	2022	1993	29	1240	−3,013,348	10.594	0.525
single-node CPU	32	1	1110	1093	29	1240	−3,013,348	10.594	0.525
multi-node CPU	16	1	2246	2204	29	1240	−3,013,348	10.594	0.525
multi-node CPU	32	2	1251	1227	29	1240	−3,013,348	10.594	0.525
multi-node CPU	64	4	769	754	29	1240	−3,013,348	10.594	0.525
multi-node CPU	128	8	498	488	29	1240	−3,013,348	10.594	0.525

The distributed layout measurements in Table 26 identify one observed limit on ideal scaling. The maximum local overlap vector shrinks with rank count, but the sum of rank-local overlap DOFs increases from about the owned problem size on one rank to more than four times the owned size on 128 ranks. This overlap replication comes from subdomain boundaries and high-order element support, so it is a decomposition cost rather than duplicated global mesh storage.

Table 26: Distributed layout measurements for the converged Plasticity3D scaling run. "Max local DOFs" is the largest local-overlap vector size on any rank; "Overlap / owned DOFs" is the summed rank-local overlap DOF count divided by the global owned free-DOF count.

CPU setting	Ranks	Nodes	Max local DOFs	Overlap / owned DOFs
single-node CPU	1	1	4,859,595	1.01
single-node CPU	2	1	2,495,265	1.05
single-node CPU	4	1	1,284,351	1.11
single-node CPU	8	1	669,783	1.21
single-node CPU	16	1	351,219	1.41
single-node CPU	32	1	182,613	1.83
multi-node CPU	16	1	351,219	1.41
multi-node CPU	32	2	182,613	1.83
multi-node CPU	64	4	91,821	2.66
multi-node CPU	128	8	50,637	4.31

7.8 Topology Optimization

The reduced design problem and objective history are defined in Section 5.6, with mechanics subproblem (35) and reduced objective (36). The numerical question here is distributed outer-loop stability: the mechanics solve and the design update must remain synchronized under continuation in p and move-regularized updates. Because the outer-loop work can vary with the continuation schedule, we report end-to-end timing under adaptive stopping rather than strong scaling at a fixed computational workload.

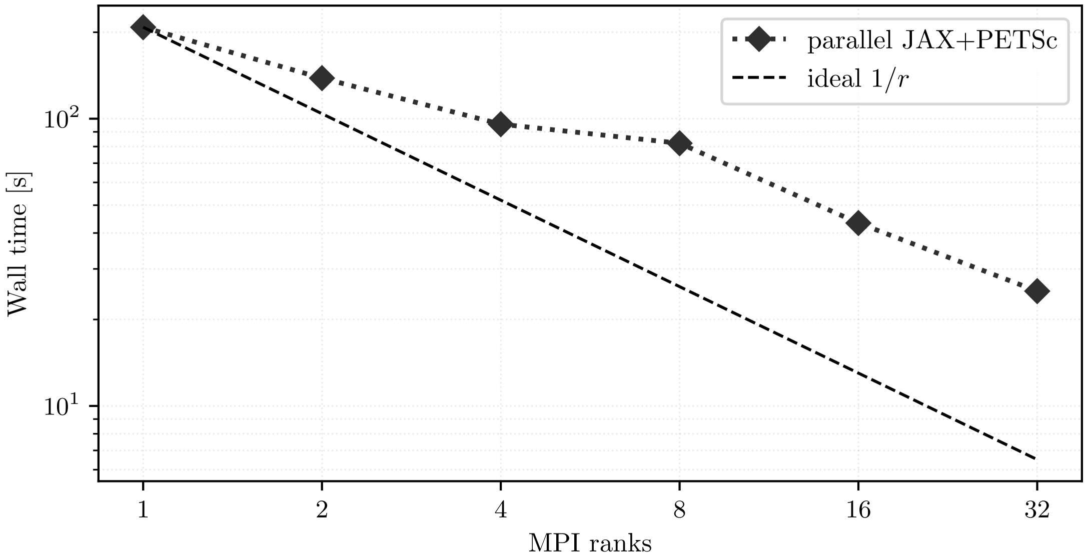


Figure 29: Topology end-to-end parallel timing trend for the fine-grid run. The plotted wall times correspond to the coupled reduced-design problem defined by (36), with per-rank stopping behavior reported in table 27.

Table 27: Parallel topology endpoint observables and timing for the fine-grid reduced-design study.

Ranks	Wall time [s]	Solve time [s]	Outer iters	p	Compliance	Volume fraction
1	208	200	65	4.2	9.1559	0.3884
2	138	134	72	5.6	8.9473	0.3932
4	95.7	92.8	69	4.6	9.1684	0.385
8	82.1	80.1	67	7	9.2178	0.3932
16	43.3	41.9	66	5.8	9.6859	0.3798
32	25.1	23.7	66	6.6	9.9074	0.3749

Figure 29 and table 27 report that the fine-grid end-to-end wall time falls from 208 s on one rank to 25.1 s on 32 ranks. The outer iteration counts and final continuation exponents vary across ranks, and the adaptive endpoints have compliance values from 8.9473 to 9.9074 and volume fractions from 0.3749 to 0.3932. These are achieved volumes under the adaptive multiplier and stopping rule, not equality-satisfaction claims for the target volume. We therefore measure adaptive end-to-end behavior for the coupled design-mechanics computation, not fixed-work strong scaling or a rank-invariant endpoint.

Table 28 complements the end-to-end timing table with a controlled 40-iteration rank-consistency test on a smaller 384×192 grid. Early convergence and stall stopping are disabled, and all rank counts use the same initial design and smoother continuation schedule ($p \leftarrow p + 0.1$). The final density field is compared with the one-rank JAX+PETSc baseline for this controlled run after 40 outer iterations. All rank counts complete the same 40 outer iterations. The compliance difference stays below 0.2%, and the density relative L^2 difference remains below 0.04, which checks that the distributed design-gradient exchange and mechanics coupling remain bounded under rank variation at this resolution.

Table 28: Topology controlled 40-iteration rank-consistency check. Each rank count completes the same 40 outer iterations; compliance and density differences are relative to the one-rank case.

Ranks	Schedule	Outer	Solve [s]	Compliance	Volume	p	Rel. compliance diff.	Density rel. L^2
1	fixed schedule	40	48.3	15.56	0.3085	5	0	0
2	fixed schedule	40	28.5	15.56	0.3083	5	1.29×10^{-6}	1.37×10^{-2}
4	fixed schedule	40	19.7	15.5308	0.3086	5	1.87×10^{-3}	1.91×10^{-2}
8	fixed schedule	40	16	15.5665	0.3081	5	4.16×10^{-4}	3.68×10^{-2}
16	fixed schedule	40	8.23	15.5606	0.3084	5	4.15×10^{-5}	1.47×10^{-2}
32	fixed schedule	40	6.43	15.5637	0.3083	5	2.41×10^{-4}	9.71×10^{-3}

The topology case extends the evidence from state solves to coupled design–mechanics optimization. Here the primary distributed JAX+PETSc realization is less about exact Hessians than about stable distributed design-gradient exchange and mechanics coupling, while pure JAX remains the compact serial design demonstration discussed earlier in Table 5.

7.9 Synthesis of the Computational Evidence

The results above provide the evidence blocks used in Section 8. The globalization tests separate problem-dependent nonlinear policies; the derivative-route comparisons isolate cost under fixed terminal observables; the distribution, PMG, and partitioning studies expose solver-policy and sparse-ownership effects; and the topology study closes with an end-to-end design-and-mechanics loop plus a controlled rank-consistency check. Read together, these measurements support the paper's main message: local differentiable constitutive or energy evaluations can be coupled to PETSc-owned sparse nonlinear and linear solves across scalar PDEs, mechanics, plasticity, and topology, with comparator evidence scoped by problem family.

8 Discussion

The numerical evidence supports a methodological conclusion for the tested benchmark classes: in nonlinear finite-element energy minimization, the useful unit of algorithmic design is not the AD route alone. Local derivative construction, sparse ownership, nonlinear globalization, Krylov algebra, and preconditioning together determine the observed solver behavior. This coupling is why the paper reports formulation checks, matched terminal-state comparisons, fixed-work route studies, and distributed scaling evidence in one computational evidence narrative.

Derivative information is realized through the nonlinear solve. The Plasticity3D derivative-route ablation is the clearest fixed-work example; see Figure 25 and table 23. On the fixed $P_4(L_1)$, $\lambda_{\text{sr}} = 1.5$, 8-rank case, the three routes report terminal observables with the same rounded energy, external-work observable, and maximum displacement, with 8 Newton iterations and 376 Krylov iterations. Thus the reported evidence isolates derivative-route cost only after the residual, Jacobian, preconditioner, and nonlinear stopping policy have been fixed.

In this setting, constitutive AD has the lowest measured wall time, while colored SFD remains a useful sparse-recovery comparison and fallback route on the tested fixed-size problems. The conclusion is therefore not that one derivative route is uniformly best, but that derivative placement should be assessed inside the assembled sparse nonlinear solve that will use it.

Agreement evidence depends on matched definitions. The hyperelastic JAX-FEM comparison in Figure 19 and table 13 is useful because the mesh, displacement schedule, and terminal-energy post-comparison are explicitly matched. The three-dimensional plasticity comparison in Figure 20 and table 14 uses a different agreement contract: fixed-load endpoint observables satisfy the stated tolerances, while deviatoric-strain and boundary-profile comparisons serve as diagnostics. These two examples illustrate the same validation rule: external or surrogate comparisons are informative when the problem definition and compared observables are named before agreement is interpreted. The Plasticity3D result is therefore endpoint-scoped mechanics evidence, not a path-consistent reproduction of incremental-history plasticity references.

Scalability evidence reflects solver policy. The distribution and PMG studies show that sparse ownership, overlap size, coarse solves, and preconditioner choices enter the numerical method. The $\lambda_{\text{sr}} = 1.0$ three-dimensional plasticity strong-scaling study is timing-only evidence, whereas the bottom-clamped $\lambda_{\text{sr}} = 1.55$ scaling sweep is the converged benchmark comparison. The topology study reports end-to-end timing under adaptive stopping, then adds a controlled rank-consistency check to separate distributed design-gradient exchange from adaptive termination effects. The fixed-reference PMG results should similarly be read as solver-sensitivity tests, not as external baselines.

Together, these distinctions state the contribution. We present a nonlinear FEM toolset that couples JAX-based local differentiation with PETSc sparse distributed solvers, and we evaluate that coupling with comparison contracts that specify their own limits. The results support this toolset and its validation discipline for the tested benchmark classes. They do not imply that a single backend, globalization policy, preconditioner, or derivative construction dominates nonlinear finite-element computations in general.

9 Conclusions

We developed a JAX+PETSc toolset for nonlinear finite-element energy minimization that couples local differentiable energy models to PETSc sparse distributed nonlinear solvers. The methodology in Sections 3 and 4 connects finite-element energy notation, second-order derivative construction, sparse MPI assembly, nonlinear globalization, Krylov solvers, and preconditioner policy. Across the tested problems, the main lesson is that local AD choices and distributed nonlinear solver design must be considered together.

The validation and performance evidence in Sections 6 and 7 support this pattern on a deliberately heterogeneous benchmark suite. Scalar problems check formulation and derivative agreement, hyperelasticity provides a narrow matched terminal-state companion comparison, three-dimensional plasticity tests nonsmooth constitutive structure and distributed sparse solves under an endpoint-surrogate interpretation, and topology optimization exercises a coupled design-and-mechanics problem. The high-order plasticity derivative-route comparison identifies constitutive AD as the lowest measured wall-time route among the tested alternatives. The converged three-dimensional plasticity scaling sweep preserves the same nonlinear and Krylov work across the tested rank counts. The topology runs provide distributed timing evidence for the coupled design-and-mechanics loop, with a separate controlled rank-consistency check.

The resulting contribution is a scientific computing framework for asking which derivative route, sparse ownership pattern, globalization policy, and preconditioner should be used together for a given nonlinear energy problem. Natural extensions are path-dependent mechanics validation against incremental-history references, broader agreement-focused external comparisons under matched problem definitions, and reduced preconditioner-setup and multigrid costs in sparse mechanics runs. Those extensions would test whether the validation discipline of the present study carries over to broader nonlinear mechanics and design problems.

A Solver-Policy Diagnostics

This appendix collects supporting solver-policy diagnostics that are methodologically useful but narrower than the main benchmark narratives.

A.1 Plasticity3D Reference-Formula and Fixed-Reference PMG Comparisons

The primary Plasticity3D numerical story in Section 7.7 is the rank-local constitutive-AD path. The tables below add two solver-policy comparisons on the same $P_4(L_2)$ problem from (32) at $\lambda_{\mathrm{sr}} = 1.0$: a closed-form constitutive-tangent assembly comparator and a frozen-preconditioner PMG variant. Both use Armijo Newton, a gradient-only stopping target $\|\nabla \mathcal{F}\|_2 < 0.01$, at most 50 Newton iterations, and PETSc relative KSP tolerance 1×10^{-1}. Here closed-form assembly means evaluating the active constitutive branch and tangent formulas without AD, while fixed-reference PMG means a frozen-preconditioner variant in which the multigrid operator is not rebuilt along the nonlinear path.

Table 29: Matched constitutive-assembly versus reference-formula assembly comparison for the same PMG solver configuration on $P_4(L_2)$ at $\lambda_{\mathrm{sr}} = 1.0$. The table reports wall time, solve time, and their ratio for matched rank counts on (32).

Ranks	AD wall [s]	Closed-form wall [s]	AD solve [s]	Closed-form solve [s]	Wall ratio
4	1717	951	1634	890	1.805
8	1164	697	1107	656	1.671
16	608	374	575	350	1.628
32	300	277	284	260	1.085

These diagnostics support two narrower statements. First, closed-form branch tangent assembly has lower reported wall time than AD-branch tangent assembly in the listed matched-rank comparisons, although the gap narrows at higher rank counts. Second, the fixed-reference PMG profile is not the default used in the present study: it may appear favorable on completed low-rank configurations, but three of the eight fixed-reference rows reach the maximum number of Newton iterations and therefore give less consistent convergence evidence. The rank-local constitutive-AD path remains the primary Plasticity3D realization because it uses the same local differentiable formulation as the derivative-route and converged scaling studies; the reference-formula configurations are retained to expose solver-policy sensitivity on matched discrete problems.

Table 30: Fixed-reference PMG alternative on the same $P_4(L_2)$, $\lambda_{\mathrm{sr}} = 1.0$ Plasticity3D problem. The status column distinguishes completed configurations from rank counts that reach the maximum number of Newton iterations before satisfying the target gradient tolerance.

Outcome and wall time

Ranks	PMG operator and tangent	Status	Wall time [s]
4	frozen PMG operator, AD branch tangent	completed	2116
4	frozen PMG operator, closed-form branch tangent	completed	988
8	frozen PMG operator, AD branch tangent	completed	1874
8	frozen PMG operator, closed-form branch tangent	completed	898
16	frozen PMG operator, AD branch tangent	iteration cap	1304
16	frozen PMG operator, closed-form branch tangent	iteration cap	601
32	frozen PMG operator, AD branch tangent	completed	348
32	frozen PMG operator, closed-form branch tangent	iteration cap	398

Newton–Krylov work

Ranks	PMG operator and tangent	Newton iters	Krylov iters	Final gradient norm
4	frozen PMG operator, AD branch tangent	24	204	6.53×10^{-3}
4	frozen PMG operator, closed-form branch tangent	21	210	2.49×10^{-3}
8	frozen PMG operator, AD branch tangent	33	238	2.53×10^{-3}
8	frozen PMG operator, closed-form branch tangent	27	246	9.53×10^{-3}
16	frozen PMG operator, AD branch tangent	50	196	2.21×10^{2}
16	frozen PMG operator, closed-form branch tangent	50	177	7.35×10^{2}
32	frozen PMG operator, AD branch tangent	29	250	2.24×10^{-3}
32	frozen PMG operator, closed-form branch tangent	50	193	8.25×10^{1}

A.2 Plasticity2D Reference Continuation Evidence

For Plasticity2D, the reference-continuation comparison is not the primary method of the paper, but it is still useful as fixed-policy rank and timing evidence for a fixed same-mesh PMG policy under the same outer continuation logic.

Table 31: Reference continuation with the fixed PMG policy at 8 and 32 ranks. This is supporting solver evidence for a reference continuation setting, not a primary solver comparison.

Policy	Ranks	Wall time [s]	Init Krylov iters	Continuation Krylov iters	Final λ_{sr}
fixed PMG policy	8	489	58	733	1.568403
fixed PMG policy	32	198	60	829	1.568334

The two rows show fixed-policy timing evidence rather than a validation baseline: increasing from 8 to 32 ranks reduces the reported wall time from 489 s to 198 s, while the final strength-reduction values remain close at 1.568403 and 1.568334.

Code and Data Availability

The source code, input data, processed benchmark summaries, and figure and table generation scripts supporting this study are available at `https://github.com/Beremi/jax_petsc_nonlinear_energies`.

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